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To cite this article: Zhengrui Tao , Mirko Sinico , Bey Vrancken & Wim Dewulf (2025) High-fidelity surrogate modelling for geometric deviation prediction in laser powder bed fusion using in-process monitoring data, *Virtual and Physical Prototyping*, 20:1, e2523550, DOI: [10.1080/17452759.2025.2523550](https://doi.org/10.1080/17452759.2025.2523550)

To link to this article: <https://doi.org/10.1080/17452759.2025.2523550>



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Published online: 04 Jul 2025.



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High-fidelity surrogate modelling for geometric deviation prediction in laser powder bed fusion using in-process monitoring data

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ABSTRACT

Surface defects in metallic laser powder bed fusion, including dross, staircase, and edge effects, undermine internal channel integrity, highlighting the need for as-built qualification to enable correction. Existing multiphysics simulations and trial-and-error experiments are time-intensive and inflexible for process control. This study targets actual-to-nominal errors within dimensional tolerance, proposing a high-fidelity surrogate model to predict deviations using melt pool monitoring data. Process signatures, including shape and thermal-related gray value features, were mapped onto CAD geometries. To enable localised prediction, feature volumes were divided into cuboids spanning multiple scan tracks and layers, capturing inter-hatch and inter-layer interactions. Deviation labels quantified via X-ray computed tomography allowed supervised training. The surrogate integrates a convolutional neural network (CNN) to capture spatial dependencies across scan tracks and a Transformer encoder to model inter-layer remelting. This configuration achieves root mean squared errors of 6.1 μm (11.3% of the 54.1 μm average ground-truth deviation) for up-skin and 27.4 μm (9.4% of 291.7 μm) for down-skin surfaces, with R^2 of 0.91 and 0.95, respectively. Ablation studies confirm CNN's intra-layer and Transformer's inter-layer contributions. Visualisation techniques (Grad-CAM and attention maps) highlight critical regions and layers, enhancing model interpretability and demonstrating its potential for in-situ surface quality control.

ARTICLE HISTORY

Received 6 May 2025
Accepted 17 June 2025

KEYWORDS

In-process monitoring; X-ray computed tomography; Laser powder bed fusion; Machine learning; Quality control

1. Introduction

Metallic laser powder bed fusion (PBF-LB/M) has transformed manufacturing by enabling rapid prototyping and the production of complex structures, such as conformal cooling channels [1,2] and pin-fin coolers [3]. These designs improve coolant flow, reduce pressure loss, and enhance thermal efficiency. However, significant surface deviations frequently arise in internal geometries, adversely affecting heat transfer and fluid dynamics. While standards like VDI 3405 Part 3 [4] recommend positioning and supports to minimise deviations, these strategies have limited effectiveness for complex, enclosed channels and cavities [5].

Surface deviations in PBF-LB/M arise from multiple factors, including the staircase effect on angled surfaces, edge effects on up-skin areas, and dross or sag on down-skin surfaces [6–8]. The staircase effect results from layer-by-layer slicing discretization [9], while edge effects occur due to interactions among surface tension, scanning strategy, and material evaporation [10]. Down-

skin deviations are driven by localised heat buildup, causing over-melting and powder sintering [8,11]. A comprehensive review by Du et al. [12] emphasised that surface-connected porosity and dross formation are closely linked to inadequate melt pool support in the overhang and down-skin regions, particularly under conditions of poor heat dissipation or excessive spatter accumulation. External deviations beyond tolerance can be corrected through milling or grinding, but internal surfaces remain largely inaccessible. Design modifications that replace circular features with tear-drop shapes can mitigate deviations; however, their effectiveness heavily depends on material and process conditions [13,14]. Previous studies have explored the effects of PBF-LB/M parameters (e.g. laser power and scan speed) on surface deviations [15,16], but directly linking these parameters to deviations remains challenging. Alternatively, offline multiphysics simulations and trial-and-error experiments can guide design and process tuning but are costly, time-intensive, and lack responsiveness to in-situ building status.

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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/17452759.2025.2523550>

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In summary, surface deviations in PBF-LB/M arise mainly from staircase formation, edge effects, down-skin dross, and powder sintering on side surfaces. These deviations reduce dimensional accuracy and impair thermal-fluidic performance, particularly in enclosed features where post-processing is impractical. Accurate prediction is thus essential for quality assurance and the advancement of qualify-as-you-build strategies.

To address these challenges, in-process monitoring and feedback control enable flexible adjustments of process parameters near feature boundaries based on predicted geometric deviations, as illustrated in Figure 1. This study aims to correlate process signatures with geometric deviations by developing a high-fidelity surrogate model.

In-process monitoring provides real-time building status, supports process control, and generates digital certificates for quality assurance and traceability. Moshiri et al. [17] demonstrated that near-infrared photodiodes effectively detect and quantify dross formation through hotspot analysis around overhangs. DePond et al. [18] employed spectral-domain optical coherence tomography to quantify layerwise roughness in PBF-LB/M-fabricated SS316L, analysing the effects of spatter, scan rotation, and laser parameters. Fleming

et al. [19] implemented inline coherent imaging for in-situ surface morphology tracking, utilising laser ablation to remove material accumulations and refill depressions, thereby significantly reducing surface roughness. Bhatt et al. [20] utilised synchrotron radiation imaging to investigate Ti-6Al-4 V parts' top and side surface roughness. A surface topology matrix was developed to characterise surface features and predict defects, which were further validated via X-ray computed tomography (XCT). Melt pool monitoring has become an integral tool for real-time quality assessment in PBF-LB/M, offering spatiotemporal signatures for detecting anomalies during fabrication [21].

Machine learning (ML) and deep learning (DL) have further advanced in-process surface quality monitoring. Knaak et al. [22] employed convolutional neural networks (CNNs) with high-dynamic-range (HDR) imaging to classify Inconel 718 surface roughness with >80% accuracy across five S_a -based roughness classes. Their findings indicate that a minimum image resolution of $5.66 \mu\text{m}/\text{pixel}$ is necessary to attain this level of classification performance. Zhang et al. [23] improved the height topography measurement in PBF-LB/M by enhancing fringe projection profilometry (FPP) through the integration of HDR imaging and DL techniques. By employing a denoising diffusion probabilistic model, the method reduced the root mean squared error (RMSE) from $10.57 \mu\text{m}$ to $4.35 \mu\text{m}$. It significantly enhanced spatial resolution, improving from $38 \mu\text{m}$ to $5 \mu\text{m}$ laterally and from $10 \mu\text{m}$ to $1 \mu\text{m}$ vertically. In a recent study, Zhang et al. [24] correlated spatters with layerwise surface roughness by combining FPP with off-axis camera imaging. Melt pool spatter characteristics, including spatter count and ejection angle, were extracted and used as input features in a support vector machine (SVM) model to predict surface roughness. The inclusion of spatter-related metrics substantially improved prediction accuracy, reducing the root mean squared error (RMSE) from $5.0 \mu\text{m}$ to $2.2 \mu\text{m}$, which is an improvement of over 50% compared to conventional models based solely on nominal process parameters without in-process monitoring. Gerdes et al. [25] used a CNN combined with cell-scan hyperspectral imaging to predict the R_z surface roughness of PBF-LB/M-fabricated magnesium alloy, achieving a mean absolute error (MAE) of $4.1 \mu\text{m}$ across samples produced under three distinct parameter sets. Given that this error is substantially lower than the observed variability in measured roughness ($\text{MAE} = 14.3 \mu\text{m}$), the model demonstrates high predictive fidelity. Toyserkani et al. [26] utilised six ML algorithms to classify the top-layer surface roughness of PBF-LB/M cuboids from co-axial photodiode data. The surface was divided into $(690 \times$

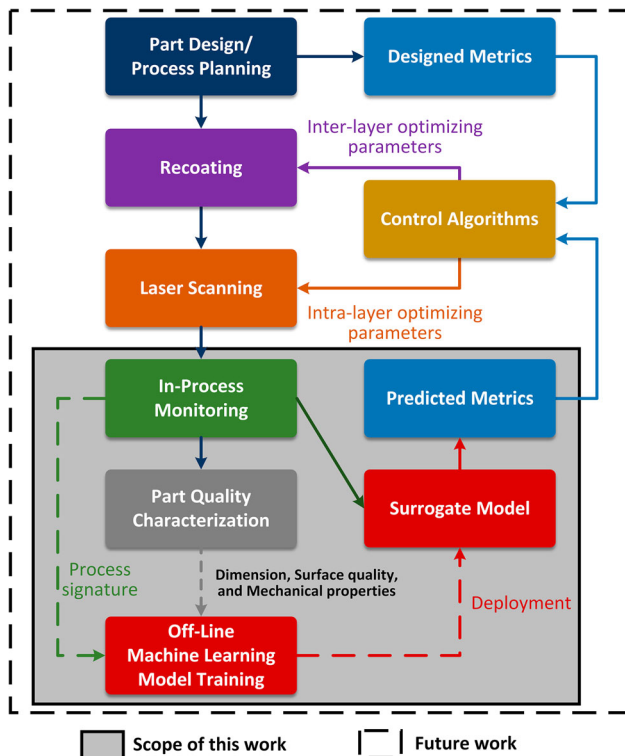


Figure 1. Framework of in-process monitoring and feedback loop control, highlighting the development of a surrogate model for quality assurance using in-process monitoring data.

510) μm^2 patches and categorised into four S_a -based roughness classes. Random Forest achieved a micro-average area under the Receiver Operating Characteristic Curve of 0.89 and a weighted F1-Score of 0.71. Akhil et al. [27] characterised the surface quality of Ti-6Al-4V components using image-based texture features. Key first- and second-order statistical descriptors were selected using neighbourhood component analysis and utilised to train machine learning (ML) models for roughness prediction. The Gaussian process regression model demonstrated superior performance, achieving an R^2 value exceeding 0.9.

Despite these advancements, most studies focused on top and side surfaces, while down-skin surfaces facing the powder bed remain under-researched due to their lack of direct observability. Bugatti et al. [28] developed a second-order linear model correlating the time above a thermal threshold (extracted from high-speed thermal video) with S_a values for 15°, 30°, 45°, 50° overhang surfaces, achieving an RMSE of 4 μm on SS316L and AISi10Mg under varying process conditions. Sen et al. [29] used optical tomography (OT) to estimate overhanging surface roughness by extracting layerwise brightness variations as predictors and fitting them with quadratic and linear models. However, both studies rely on limited datasets with restricted coverage of overhang angles and focus on externally accessible surfaces measurable by profilometers. Correlating process signatures with internal surface deviations thus remains a critical gap. In addition, high-quality in-process monitoring data are essential for extracting physically informed features and constructing accurate labelled datasets. These features should closely reflect process stability and the underlying mechanisms of defect formation and evolution [30]. Considering intra-layer and inter-layer remelting effects is also critical for developing high-fidelity surrogate models [31].

This study develops a data-driven surrogate model for predicting local surface deviations, focusing on the horizontally built channels and 45° overhang surfaces from the NIST open dataset 'Overhang Part X4' [32,33], where down-skin and up-skin inner surfaces exhibit average actual-to-nominal deviations of 191.7 and 54.1 μm , respectively. The key contributions include:

- (1) Incorporating Geometric Context into Process Signature Mapping: Shape [34] and thermal-related gray value features [28] were extracted from co-axial melt pool monitoring (MPM) images. These process signatures were mapped onto the CAD geometry based on the laser spot position to align melt pool behaviour with the geometric design intent. This

mapping allows the surrogate model to consider critical geometric features (e.g. edges, corners, and overhangs) when predicting surface deviations. Feature volumes were resampled into smaller cuboids at boundaries, covering multiple scan tracks and layers as model inputs. Besides, actual-to-nominal surface deviation labels were quantified using XCT, providing high-accuracy internal surface measurements and overcoming the inaccessibility of surface profilometry [35].

- (2) Hybrid CNN-Transformer Architecture for Spatial-Temporal Modelling: The proposed surrogate model integrates a CNN and a Transformer encoder to capture multi-scale spatio-temporal dependencies within sampling cuboids. The CNN acts as an automatic feature extractor, hierarchically distilling geometric and thermal features across scan tracks while capturing inter-hatch effects within each layer. The Transformer encoder leverages self-attention mechanisms to model inter-layer interactions, contextualising heat and material flow dynamics, including remelting and heat accumulation. This integration enables accurate surface deviation predictions by connecting initial layer conditions with subsequent thermal cycles.
- (3) Model Generalizability and Interpretability: The model's generalizability was validated on internal channel geometries (up-skin and down-skin surfaces) and external overhang surfaces. Ablation studies highlighted the distinct contributions of the CNN in feature extraction and the Transformer in inter-layer interaction modelling. Interpretability techniques, including gradient-weighted class activation mapping (Grad-CAM) and attention visualisation [36,37], provided insights into the model's decision-making process, increasing confidence in its predictions.

The rest of the work is organised as follows: Section 2 describes the experimental design, XCT analysis, dataset assembly, and data-driven modelling. Experimental results and discussions are reported in Section 3, and Section 4 concludes the study.

2. Materials and methods

2.1. Open dataset description

The sample geometry, shown in Figure 2b, features a footprint of $(5 \times 9) \text{ mm}^2$ and a height of 5 mm, incorporating a cylindrical channel (Φ 4 mm) and a 45° overhang surface. Fabricated from nickel alloy 625 (IN625) [33,38], the part was produced using a 'contours and fill' strategy

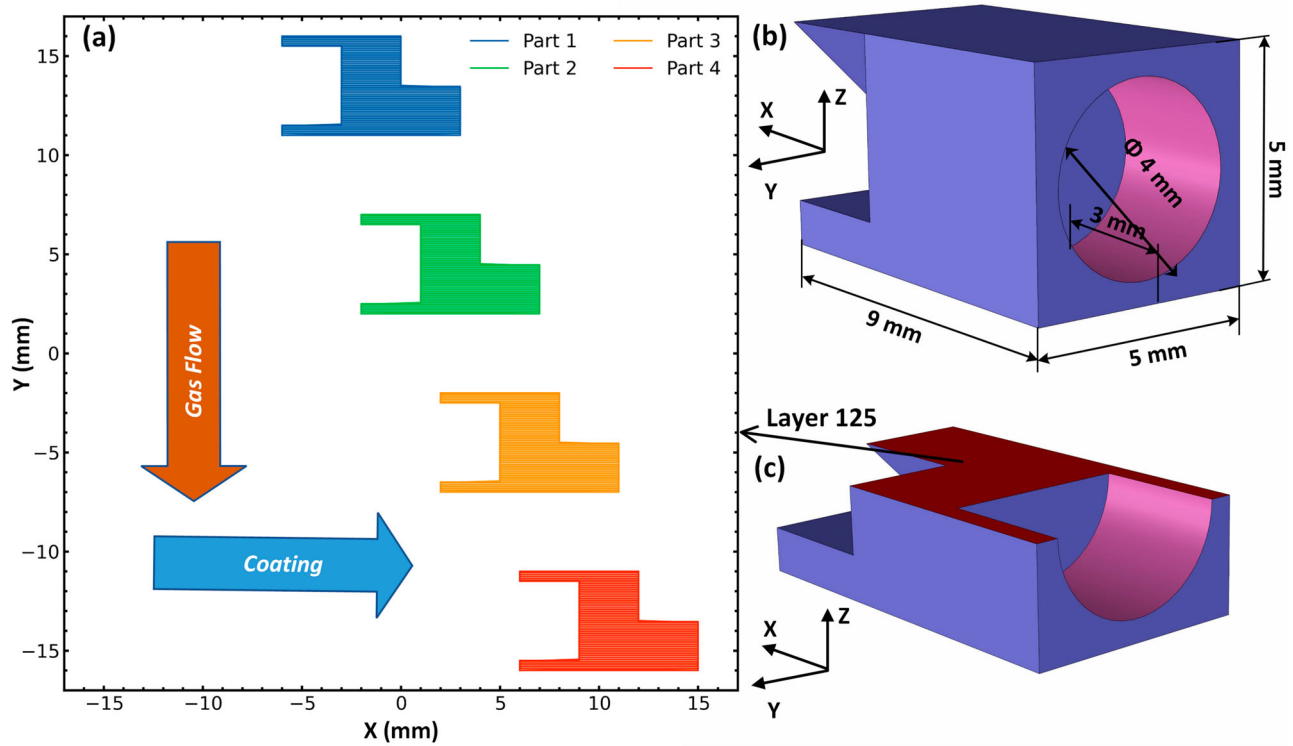


Figure 2. Part geometry design: (a) part numbering and baseplate layout, adapted from [33]; (b) part geometry; (c) cross-section of 125th layer.

with scan speeds of 900 mm/s for the contour and 800 mm/s for the infill. Because the laser closely followed the part's boundary, side surfaces were offset by approximately half the melt pool width. This study focuses on the up-skin and down-skin inner surfaces within the cylinder and the 45° external overhang plane.

A co-axial MPM system captured thermal radiation emitted from the melt pool, recording its near-infrared intensity (850 ± 20 nm) as 8-bit grayscale proxies for relative temperature. Images were acquired at 10 kHz with a 20 μ s integration time. As illustrated in Figure 3a, the optical path includes a 1070 nm fibre laser operating at 195 W with a $D4\sigma$ spot size of 60 μ m. Process emissions were reflected within the scan head and redirected by a dichroic mirror, which reflects wavelengths below 980 nm while transmitting the laser beam. These emissions were filtered and captured by a CMOS camera with an 850 ± 20 nm bandpass filter to suppress laser and ambient light. The camera was triggered by the same digital commands as the laser position and power, allowing for precise mapping of each image frame to its corresponding scan location.

Each MPM image consists of 120×120 pixels at 8 μ m/pixel resolution. MPM images were rotated 5.8° clockwise around pixel (68, 57), ensuring that the MPM coordinate system {M} is co-oriented with {A}. The MPM image can then be translated within {A} according to

the laser scan positions (Figure 3b). Additionally, both coordinate systems are discretized at 8 μ m/pixel; therefore, the MPM data can be directly stitched into the pixelated powder bed domain, as detailed in Section 2.3.2. More details about the process parameters and monitoring setup can be found in [33].

2.2. XCT surface deviation labels

The NIST-provided XCT data consisted of 16-bit grayscale TIFF slices in the Z-direction at a 12 μ m pixel size (X and Y) and a 12 μ m slice spacing (Z). More information on acquisition, reconstruction, registration, and slicing can be found in [39]. The XCT volumes were reconstructed from these slice stacks using VGSTUDIO MAX (Hexagon, SE) with a 12 μ m voxel size (Figure 4). The material boundary was determined using the ISO50 method, followed by surface refinement in 'advanced mode' (8-voxel search distance) to remove artifacts and enhance sub-voxel precision. Detachment from the substrate by wire electrical discharge machining removed support structures, reducing the part height from 5 mm to approximately 4.8 mm.

To access surface deviation, the XCT volume was aligned with the nominal CAD using VG's sequential alignment method [40]. The primary datum was selected as the top surface of the build, while the secondary and

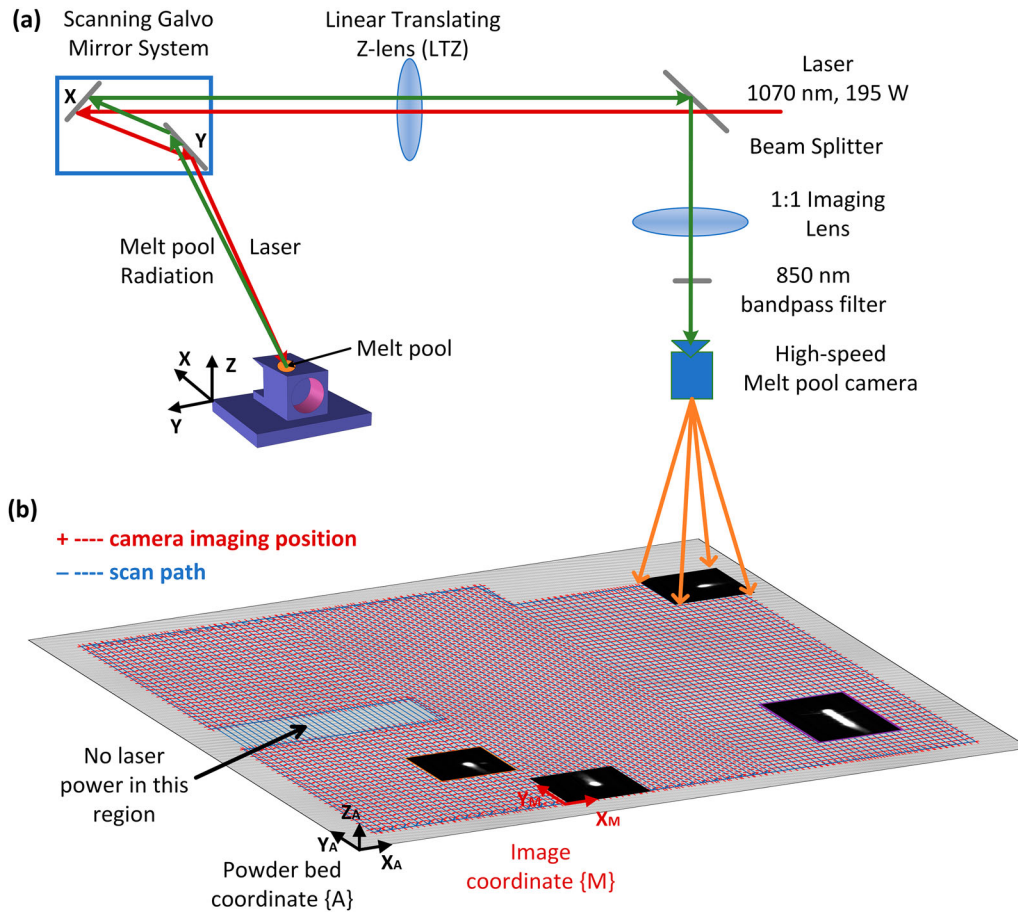


Figure 3. Experimental setup: (a) schematic of the in-process monitoring system, showing the high-speed camera co-axially aligned with the laser path, capturing process light at a wavelength of 850 nm (± 20 nm). Adapted from [33]. (b) melt pool monitoring (MPM) images were mapped back to the corresponding scan position, forming a pixelated powder bed surface, where multiple MPM images cover each pixel. This overlap enables the recording of each pixel's gray value history as a proxy for its relative thermal history.

tertiary datums were the symmetry planes of the components $-Y/+Y$ and $-X/+X$ faces, respectively. These surfaces were chosen for their large areas, which allow better averaging of small form errors. All datum surfaces were fitted using a least squares plane fitting algorithm, sequentially constraining six degrees of freedom.

Surface deviation for the 45° overhang was calculated as the normal distance from each vertex on the XCT surface to the corresponding nominal surface. Deviations on the horizontal cylinder were determined following the method of Baier et al. [14]:

- (1) Cross-sections were taken at $12\ \mu\text{m}$ intervals along a $2.52\ \text{mm}$ range centred on the cylinder axis (Figure 5a). This range starts $0.36\ \text{mm}$ from the inlet and ends $0.12\ \text{mm}$ from the outlet, excluding end zones, yielding 210 slices part. Each cross-section was cropped to retain only the actual and nominal surface profiles.
- (2) These cross-sections were converted into polar coordinates to present the 'rolled-out surface' as

horizontal curves. Thus, the arc radial deviation was converted to the distance between the curve and the nominal reference line (Figure 5b). A search algorithm identified the actual surface profile (Figure 5c), generating height maps of surface deviations (Figure 5d).

- (3) These height maps were stacked to form a 2.5D surface plot, visualising the unfolded actual-to-nominal deviation distributions along the cylinder's circumference and axis (Figure 5e). These radial deviations will label process signatures extracted from MPM images at corresponding locations.

While the XCT-measured deviation labels are considered the ground truth in this study, these labels have uncertainties that depend heavily on the specific XCT system, scan setup, reconstruction parameters, segmentation method, and registration accuracy. Providing a fully traceable uncertainty budget for these XCT measurements is currently impractical, particularly

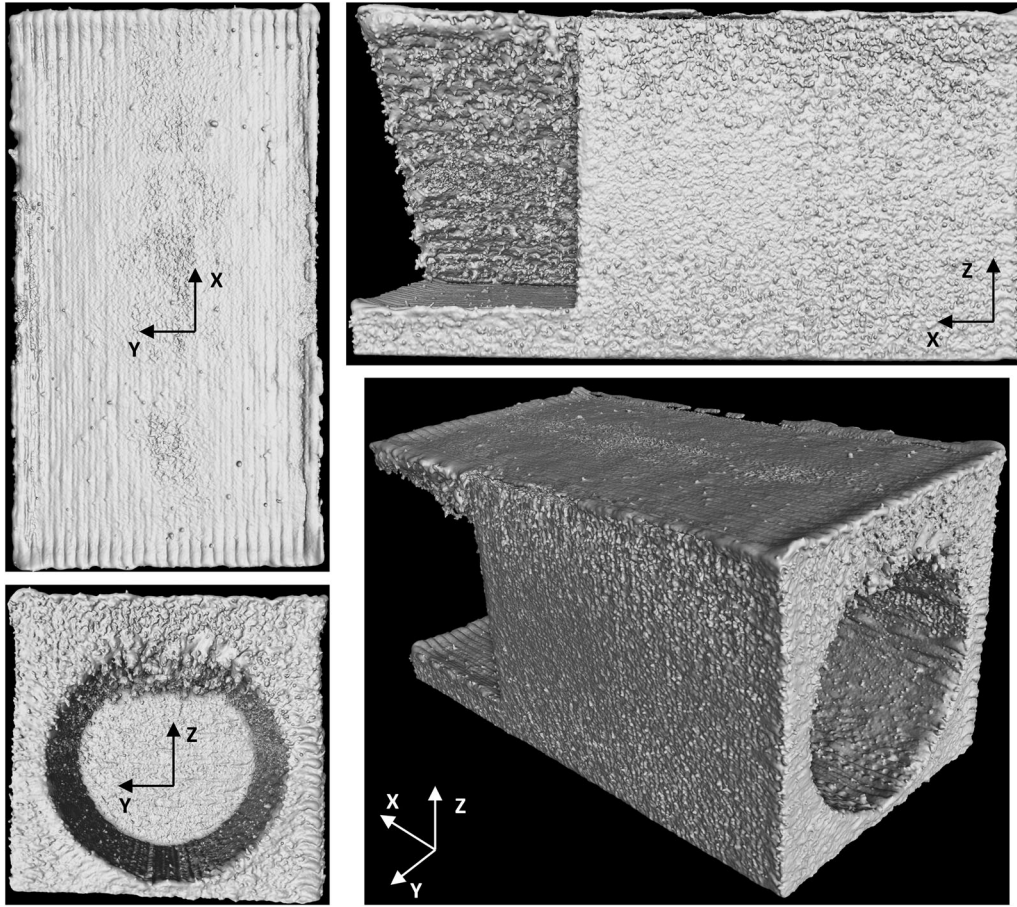


Figure 4. The top, side, and front views of the reconstructed X-ray computed tomography volume.

when the data is sourced from trusted open datasets. This study is structured by accepting the XCT data as ground truth, calculating the performance of the data-driven model against this data, and acknowledging that future studies should incorporate the uncertainty budget of XCT into the performance evaluation. You can refer to Section 3.2 for further explanation.

2.3. Feature engineering: MPM image processing

2.3.1. Low-spatial-resolution shape features

The melt pool exhibits complex size and shape variations that simple metrics (e.g. length, area) cannot fully capture [41]. This study uses directional distances from the melt pool centroid to its boundary to characterise the magnitude and directional changes, a method previously applied for clustering melt pool types [42,43]. Spatter ejections were also analysed, as they contaminate the powder bed and roughen surfaces [44,45].

Figure 6 outlines the extraction process. To extract shape features, melt pools were segmented from MPM images using a gray-value threshold of 30 out of 255 (Figure 6b) based on a nominal melt pool width of

100 μm [46]. Given their lower intensity and poor contrast, spatters were enhanced through contrast adjustment (Figure 6c) and adaptive thresholding (Figure 6d) to improve visibility. The melt pool and spatter were isolated from the background (Figure 6e) to extract six geometric features, with an ellipse fitted to the melt pool contour (Figure 6f).

The melt pool centroid O was calculated using raw moments. To standardise comparisons across moving melt pools, point A , where the scan direction intersects the melt pool boundary, serves as the starting point. From A , distances from O to boundary points were sampled at 10° increments in a clockwise direction (Figure 6g), yielding a 36-dimensional feature vector $D = \{dis_0, dis_{10}, \dots, dis_{350}\}$, defined as:

$$dis_i = \sqrt{(x_i - x_o)^2 + (y_i - y_o)^2} \quad (i = 0, \dots, 350) \quad (1)$$

where (x_i, y_i) are the coordinates of the boundary pixel at angle i and (x_o, y_o) are the centroid coordinates.

A total of 42 shape features were extracted from each MPM frame and mapped to the corresponding laser spot position, with each feature serving as a separate channel in the shape feature map. The boundary region was

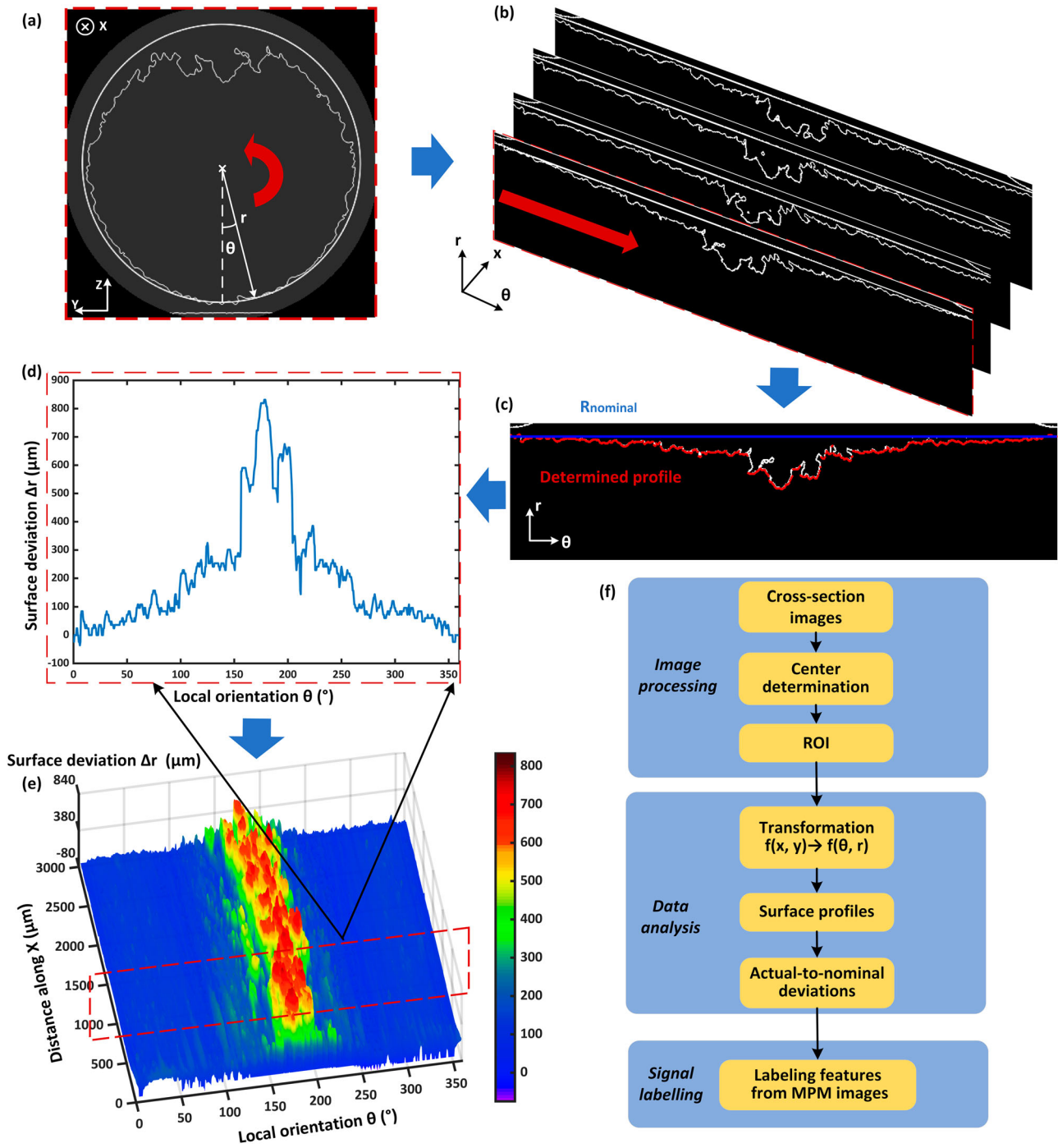


Figure 5. Surface deviation labelling: (a) cropping the region of interest (ROI) to include actual and nominal cylinder surface profiles; (b) applying a polar transformation to the ROI, (c) identifying the innermost surface pixel to determine the actual surface profile; (d) depicting surface deviation along the circumference as a height map; (e) restacking height maps along the cylinder axis; (f) overall flowchart outlining the surface deviation extraction process.

prioritised for surface deviation analysis; thus, feature maps were constructed at a $90\ \mu\text{m}$ pixel size based on the scan speed ($900\ \text{mm/s}$) and camera frame rate ($10\ \text{kHz}$). However, the $80\ \mu\text{m}$ laser spot spacing in the infilling region and $100\ \mu\text{m}$ hatch spacing introduced the rounding error, causing small blank areas ($<1\%$ per layer), as seen in Figure 6g, f.

The shape feature map was reconstructed by calculating the pixel positions of each sampling point relative to the lower-left reference near the {A} coordinate origin (Figure 3b). Distance projections along the X/Y axes were divided by $90\ \mu\text{m}$ to determine pixel positions, with each pixel storing 42 features. These maps were stacked along the Z-axis at the layer thickness of

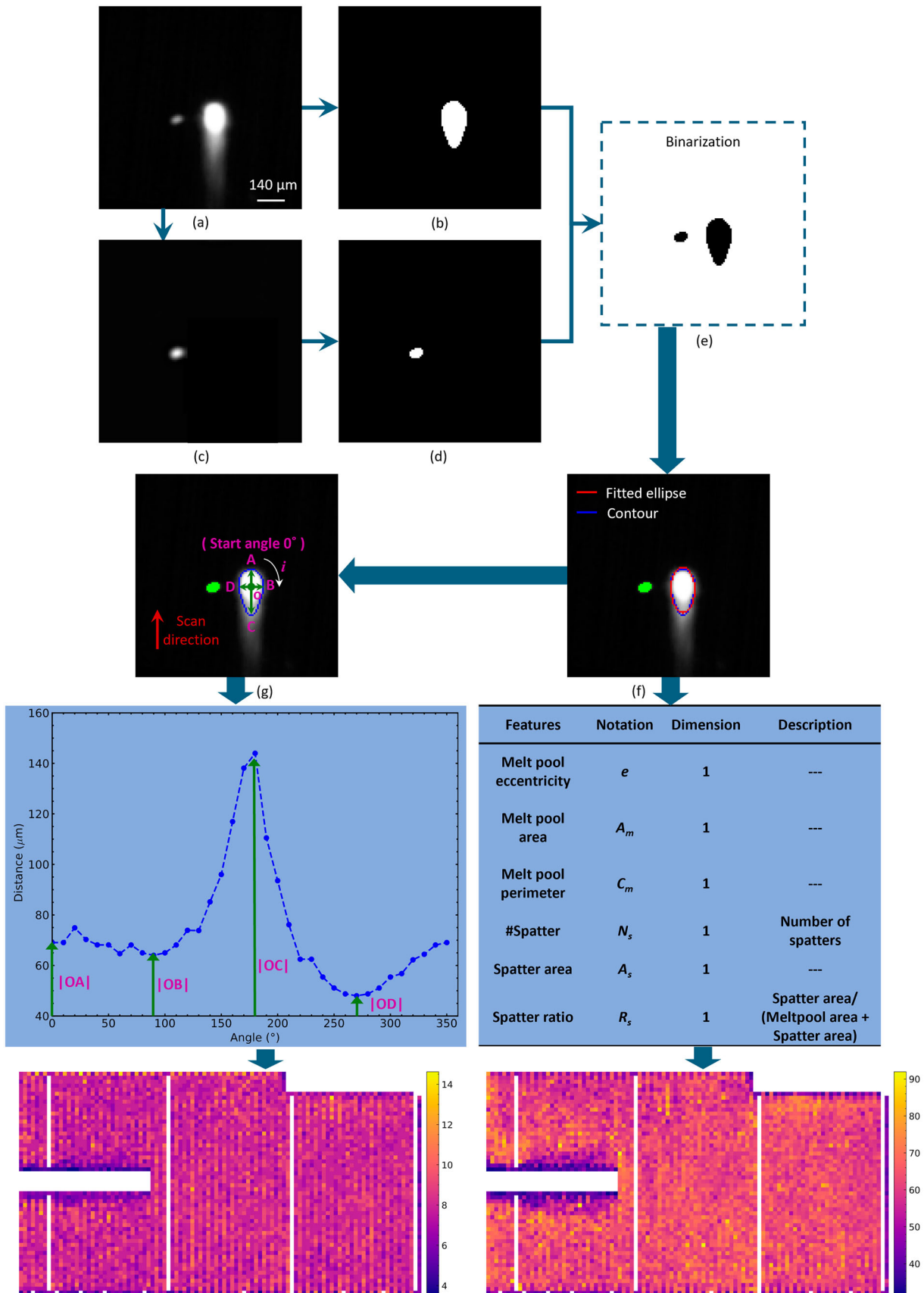


Figure 6. Procedures for extracting geometry and motion features: (a) raw melt pool image; (b) binary thresholding based on measured melt pool width; (c) and (d) spatter detection using contrast enhancement and adaptive thresholding; (e) segmented melt pool and spatter regions; (f) geometry features; (g) motion features, measured clockwise from the reference point along the melt pool boundary. Notably, some regions remain unfilled due to rounding errors, where the hatch spacing (100 μm) is not divisible by the pixel size (90 μm), causing < 1% blank areas per layer.

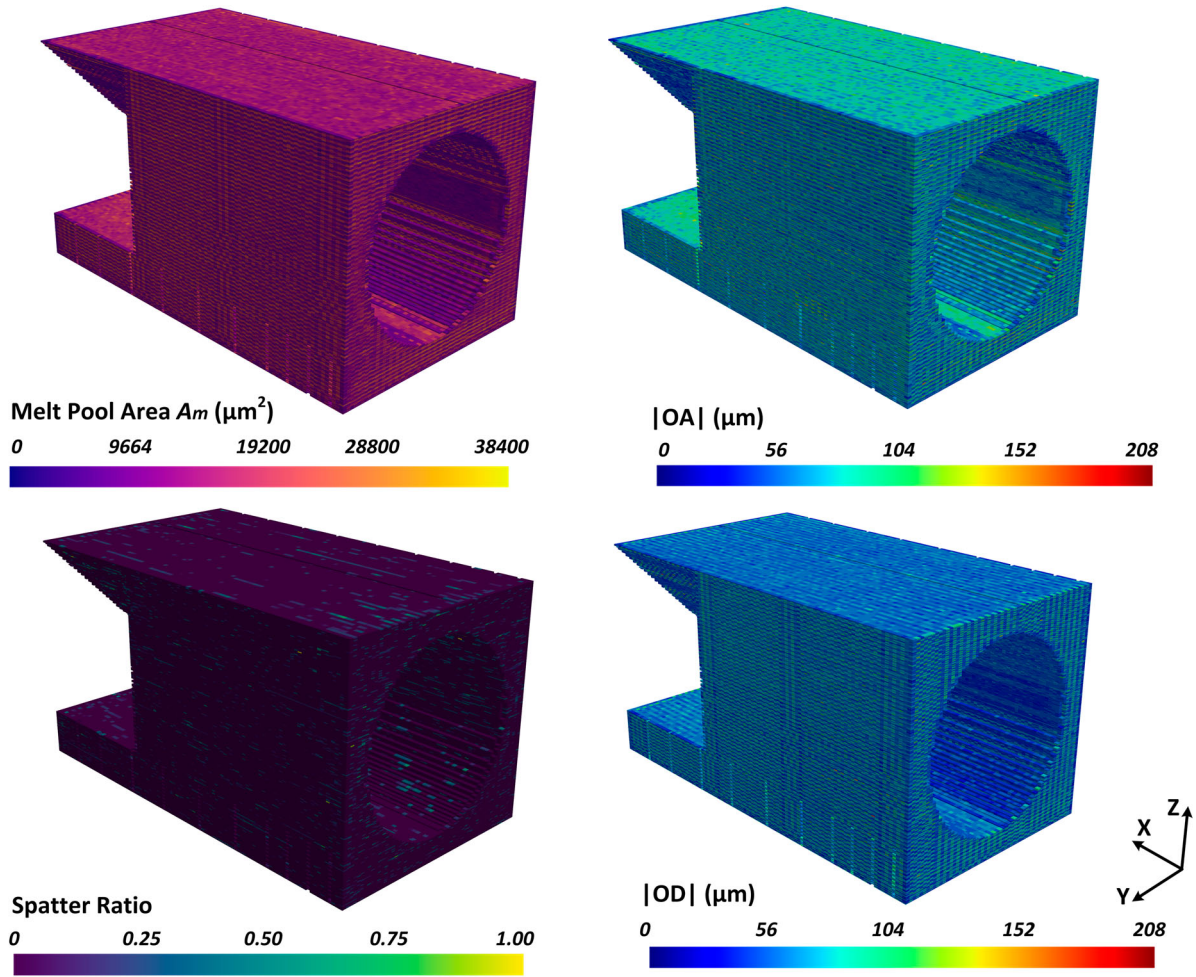


Figure 7. 3D meshes of four shape features: melt pool area, spatter ratio, and distances $|OA|$ and $|OD|$, illustrating spatial distributions of features within the reconstructed feature volume.

20 μm , forming a 3D shape feature volume. Figure 7 examples four channels of the feature volume: melt pool area (μm^2), spatter ratio (0-1), and distances $|OA|$ and $|OD|$ (μm).

2.3.2. High-spatial-resolution thermal features

MPM data were collected as a continuous image stream, with each frame stitched to the powder bed coordinate system $\{A\}$ based on laser spot positions using a superposition method [32] (Figure 8a). Because each MPM image spans a larger area than the laser movement per sampling interval, multiple frames overlap at a single pixel, creating a gray value history that reflects its thermal evolution. Unlike shape features (derived from single MPM frames), thermal-related gray value features (Figure 8b), including peak gray value, total time above a threshold, and the average level above a threshold, were extracted from this history.

Each feature was treated as a separate channel, producing a gray-value feature map (Figure 8c) that

captures thermal process signatures. A threshold of 65, corresponding to the mean melt pool intensity, was chosen to mitigate noise. Notably, the low-value regions between scan tracks were an artifact of the threshold setting rather than lack-of-fusion voids, and these regions remained connected to adjacent scan vectors.

The thermal feature maps were stacked along the build direction to form a 3D volume, as shown in Figure 9. Some voxels extended beyond the part boundary due to spatter ejections and the contour path exactly aligning with the part edge [33].

2.4. Dataset assembly

The dataset was constructed using two 3D feature volumes from section 2.3 and the 2.5D surface deviation map from section 2.2. To enable localised surface deviation prediction, the entire volume was cropped using sampling cuboids. Sampling cuboids were designed to minimise interpolation errors, with

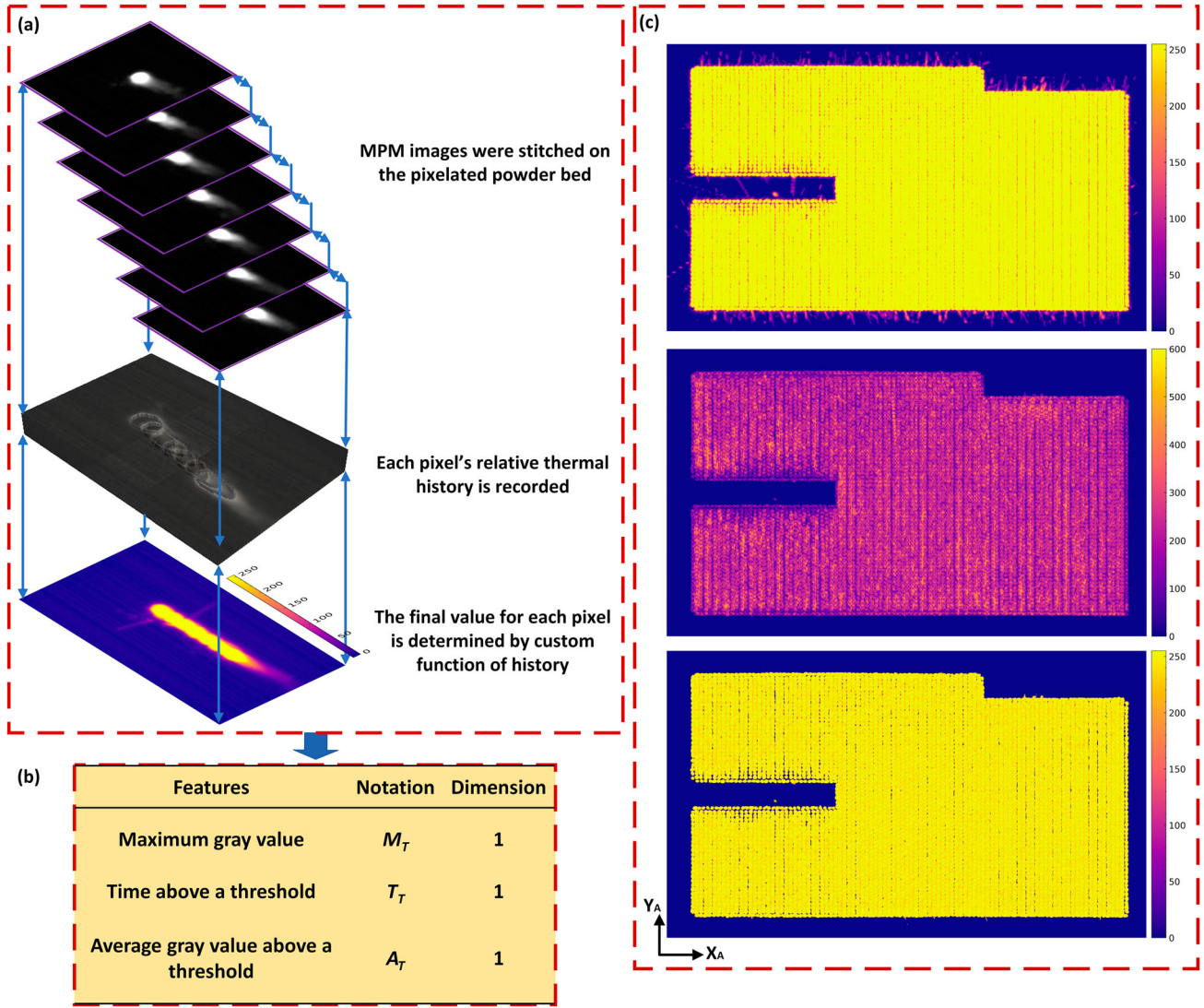


Figure 8. Procedures for extracting thermal-related gray value features: (a) image stitching algorithm where each pixel in the pixelated powder bed stores its gray value history in a vector, followed by a custom function applied to the history; (b) thermal-related gray value features; (c) high-resolution thermal feature map with three channels.

dimensions in XY plane set as multiples of the voxel size ($90 \times 90 \mu\text{m}^2$ for shape features and $8 \times 8 \mu\text{m}^2$ for thermal features), measuring $(360 \times 720) \mu\text{m}^2$ (Figure 10a). Due to heat accumulation and the laser's position inaccuracies, both the boundary and adjacent bulk regions were considered. The position of the sampling cuboid along the Y direction was determined by identifying the target layer's (0th layer) edge and extending $270 \mu\text{m}$ into the inner cylinder and $450 \mu\text{m}$ into the bulk. The range along the X direction matched the XCT labelling analysis range, spanning from 0.36 mm from the inlet to 0.12 mm from the outlet. The sampling cuboid height was set as multiples of the layer thickness to consider the inter-layer effect, resulting in dimensions (4, 8, **STEP**) and (45, 90, **STEP**), where **STEP** represents

the number of layers considered and is fine-tuned in Section 3.1.

Two feature sampling cuboids were shifted incrementally by $360 \mu\text{m}$ along the X direction, generating one data point per shift. The 0th layer, along with the arc angle and cuboid width, defined the region of interest (ROI) on the roll-out deviation map (Figure 10b). The average deviation within the ROI, corresponding to the 0th layer's arc radial deviation, served as the label for the two feature cuboids. This process produced a labelled dataset with 2800 entries per cylinder.

The four nominally identical parts in the open dataset were used to evaluate whether a surrogate model developed for one part can be effectively applied to the others. Therefore, parts 1 and 3 were

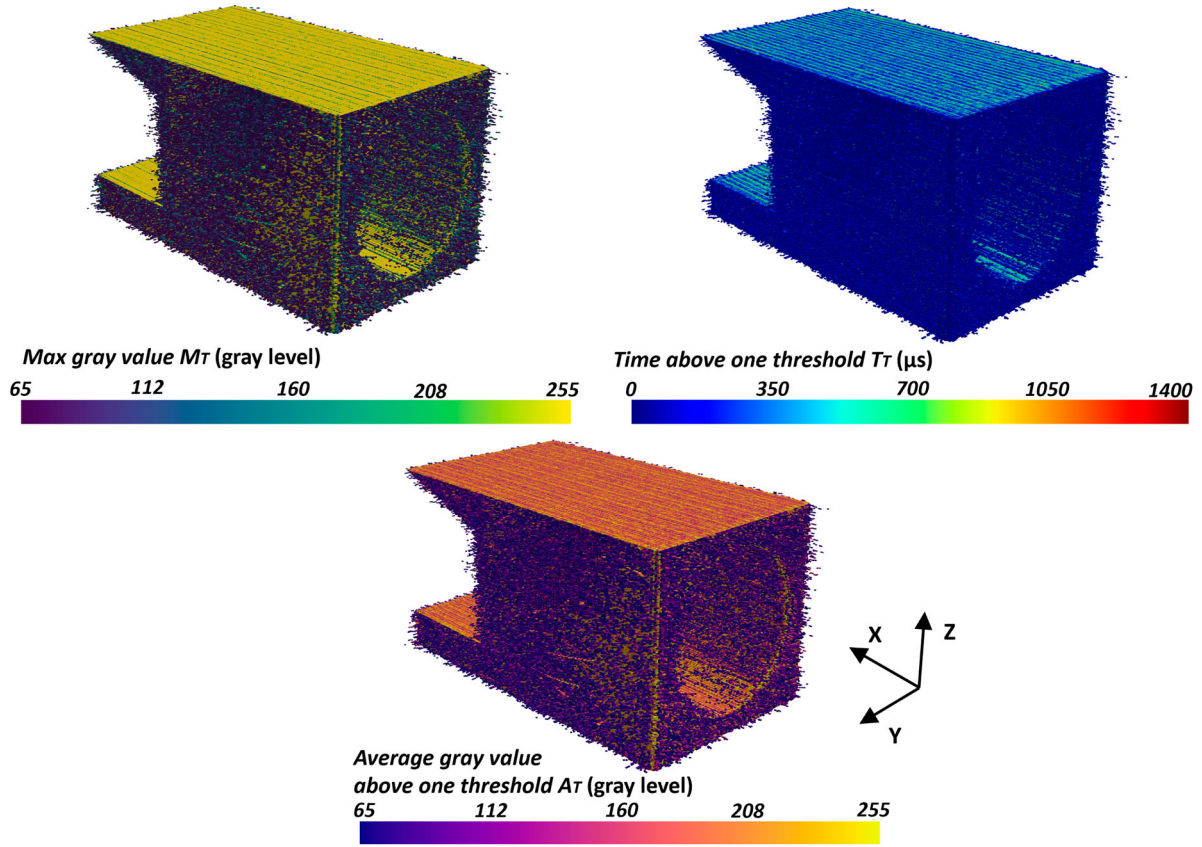


Figure 9. 3D meshes of gray-value features: max gray value, duration above one threshold, and average above one threshold, with the threshold set at 65 to reduce noise. Some voxels extended beyond the part boundary, mainly due to spatter ejections. Also, half of the melt pool lay outside the CAD geometry without a boundary offset.

used as training and validation sets, while parts 2 and 4 were reserved for testing. Although the training and testing datasets are independent, the current evaluation reflects intra-batch generalisation as four parts from the same build.

2.5. Surrogate model and performance metrics

A novel DL pipeline, combining a CNN and Transformer encoder (CaT), is proposed to capture spatial and temporal dependencies between scan tracks and printing layers (Figure 11). While the architecture shares structural similarities with dual-path video models such as SlowFast [47], it is tailored to the spatiotemporal characteristics of PBF-LB/M. It fuses low-resolution shape features and high-resolution thermal features extracted from melt pool monitoring data. Dual CNN-based feature extractors operate independently on each layer within a sampling cuboid to learn intra-layer patterns, which are then concatenated into a compact per-layer representation. A Transformer encoder then processes the sequence of layerwise representations along the build direction. Its self-attention mechanism captures

non-local dependencies across layers, enabling the CaT model to learn remelting and heat accumulation effects. Multi-head attention enhances sensitivity to diverse inter-layer interactions, while positional encoding preserves thermal propagation directionality.

Two CNN encoders, G_s and G_t , independently process the shape and gray-value cuboids, with input dimensions of $42 \times \text{STEP} \times 8 \times 4$ and $3 \times \text{STEP} \times 90 \times 45$ (channel $\times$ height $\times$ length $\times$ width), respectively. The shape cuboids were resized from 8×4 to 50×50 , as detailed in Figure 11b, before being fed into the G_s . These encoders autonomously extract hierarchical patterns using convolutional kernels, producing a 512-dimensional feature vector for each layer of the sampling cuboid. The output vectors of both CNNs were concatenated into a 1024-dimension vector. The process is repeated across **STEP** layers of a cuboid, producing a feature sequence with a length of **STEP**, as illustrated in Figure 11a–c.

Before passing the embedding sequence into the Transformer encoder, positional encoding is applied to ensure the model understands the relative position of each layer in the sequence. The Transformer encoder uses a self-attention block to capture inter-layer

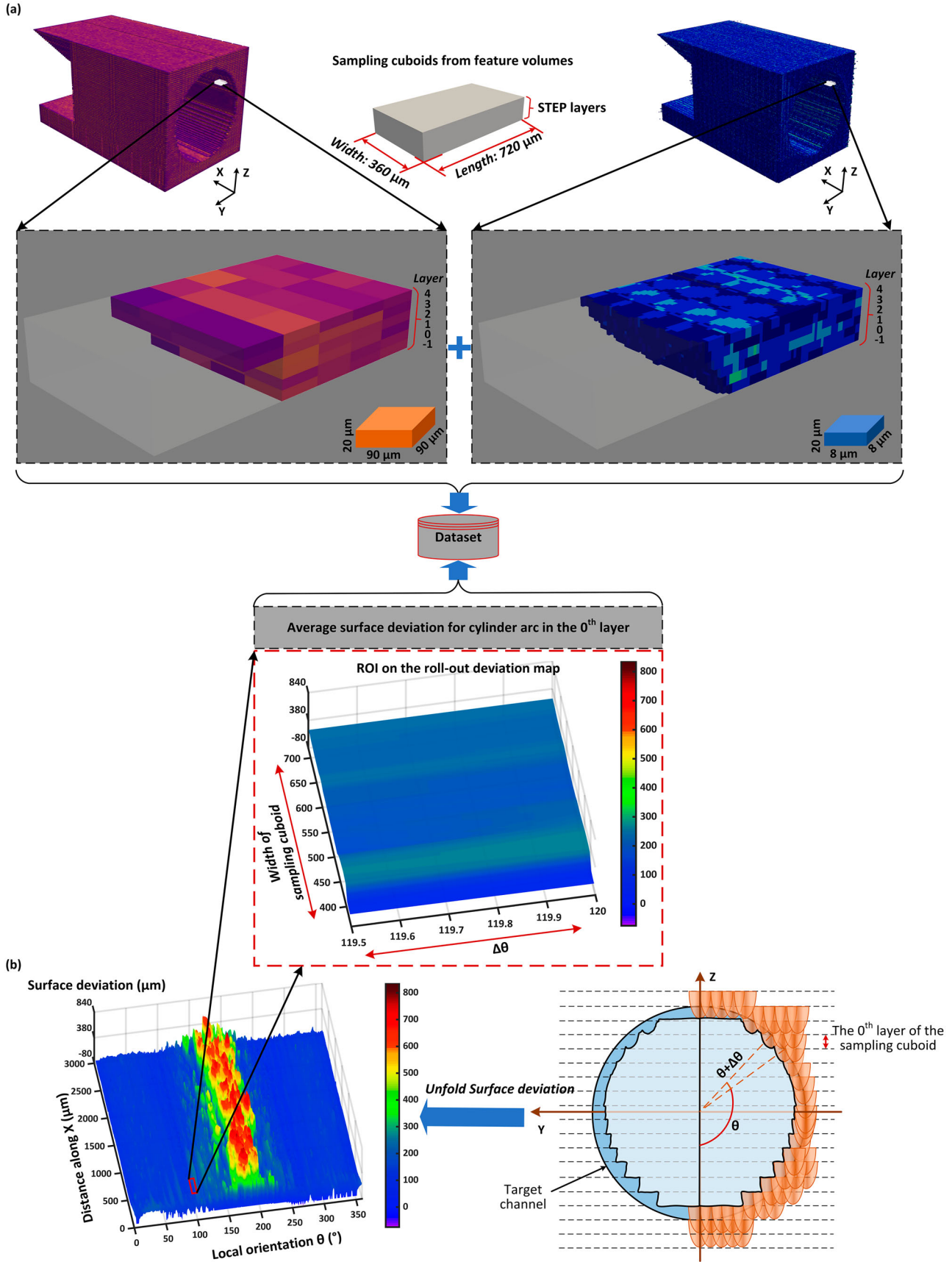


Figure 10. Dataset assembly: (a) two sampling cuboids were extracted from the 42-channel shape volume and 3-channel gray-value volume, each containing STEP layers (STEP = 6); (b) the 0th layer in the cuboid, along with the arc angle and cuboid width, determines the region of interest on the unfolded deviation map. The segmented local surface deviation was averaged to label the two feature cuboids. STEP indicates the number of layers included in the sampling cuboid.

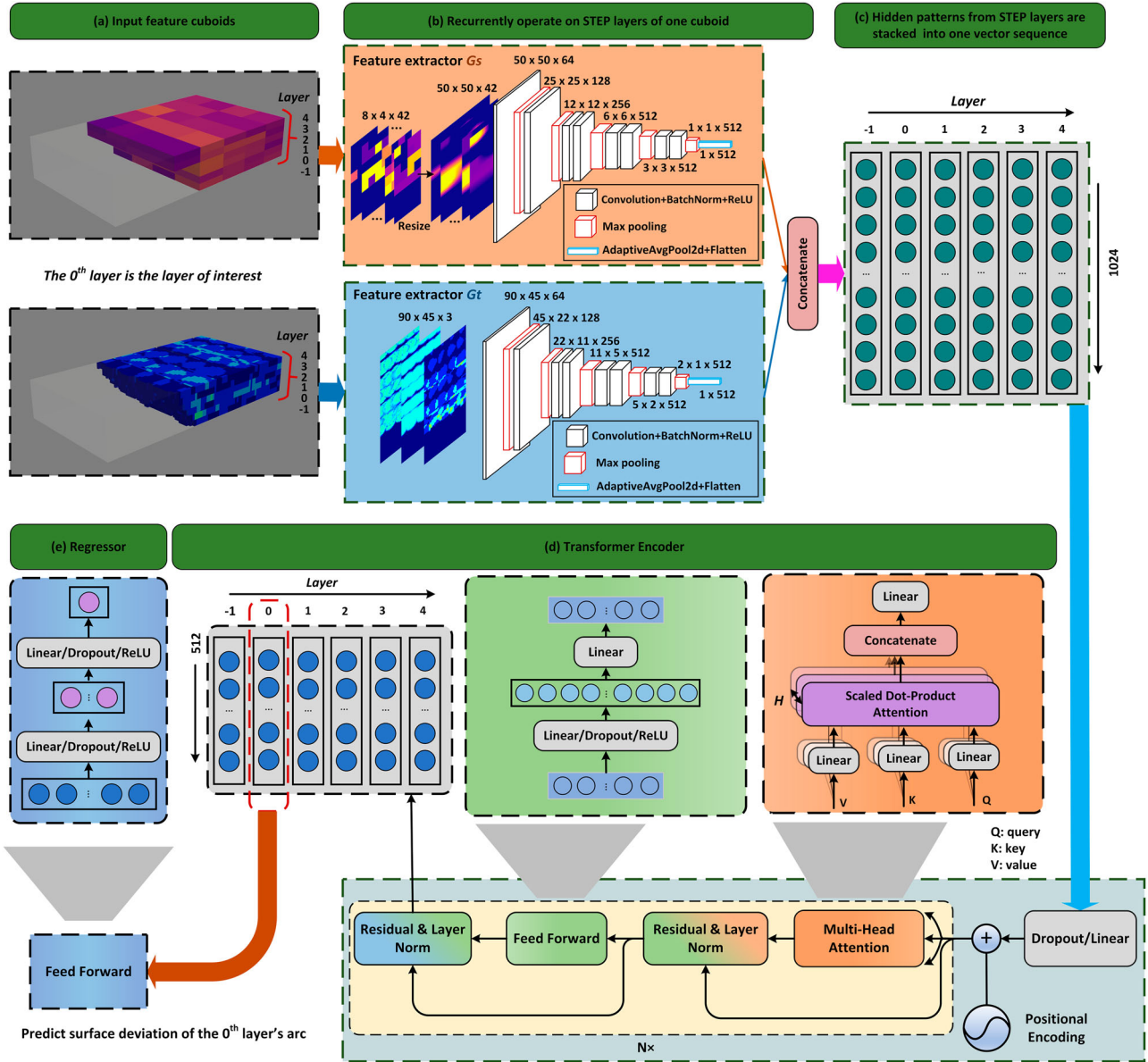


Figure 11. Prediction model: (a) feature sampling cuboids covering *STEP* layers, with *STEP* = 6 as an example, where the 0th layer is the target layer, and one layer below is considered; (b) two CNN-based feature extractors recurrently operate on each layer of a cuboid to automatically learn hidden patterns, which were joined into a 1024-element vector for each layer; (c) these vectors were stacked into a sequence of embedded features; (d) the Transformer encoder captures spatial correspondences hidden in the sequence, specifically the inter-layer effects, and outputs encodings for each sequence element; (e) the 0th layer's encoding was transformed into a prediction through a two-layer feed-forward network.

dependencies, focusing on how thermal and material flow dynamics influence the target layer's surface deviation. Attention weight matrices were learned, quantifying the influence of each layer on others. These weights enable the model to prioritise layers and features critical to predicting the target (0th) layer's surface deviation.

The Transformer encoder processes the sequence through one multi-head attention block and a feed-forward block, iteratively refining the learned representations. Residual connections and layer normalisation were integrated into each block to stabilise training

and enhance generalisation, as shown in Figure 11d. This sequence processing is repeated N times. The final encoding for the target layer is passed into a two-layer feed-forward network to predict the surface deviation for the 0th layer (Figure 11e). This design effectively combines local spatial features extracted by the CNN with inter-layer temporal dependencies captured by the Transformer encoder. Detailed CaT model configurations are provided in Appendix A.

To ensure robust performance, five key hyperparameters, including the embedded feature dimension

Table 1. Training parameters for the hybrid network combining CNN and Transformer encoder.

Training parameters	CaT network	Training parameters	CaT network
Optimiser	Adam	Batch normalisation	True
Learning rate	Cosine warm-up learning rate scheduler	Type of analysis	Regression
Total epochs	100	Loss function	Mean Squared Error
Batch size	256	Trainable weights	~13,500,000

(d_m), the number of attention blocks (N), the regressor hidden layer dimension (d_r), the number of layers below the 0th layer (l_b) in the sampling cuboid, and the total number of layers in sampling cuboid (**STEP**), were optimised using 3-fold cross-validation (CV). The training set was split into three subsets, with each potential model being trained on two subsets and validated on the third. Performance metrics, including mean absolute error (MAE) and R^2 , were averaged across folds to identify the configuration with the lowest MAE and highest R^2 . With these optimal hyperparameters, the CaT model was retrained using the whole training set. During the hyperparameter tuning and model training, the CaT model was trained for 100 epochs with a batch size of 256, using the Adam optimiser and a cosine warm-up learning rate scheduler, as specified in Table 1.

To quantify the model's predictive performance on the testing dataset, three metrics were utilised: root mean squared error (RMSE), MAE, and R^2 .

$$\left\{ \begin{array}{l} \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \\ \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\ R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \end{array} \right. \quad (2)$$

where y and $\hat{y}$ refer to the true and predicted values, respectively. $\bar{y}$ is the mean of the true values, and n is the total number of samples in the testing dataset. RMSE and MAE, measured in μm , quantify the model's prediction error. At the same time, R^2 , a dimensionless metric, represents the proportion of variance in the dependent variable explained by the independent variables, providing a percentage measure of prediction accuracy.

3. Results and discussions

3.1. Process insights: physics-driven constraints on hyperparameters

Identifying correlations between process signatures and surface deviations requires domain-specific insights. For the sampling cuboid configuration, the range for l_b and

STEP should consider the geometric relationship between the melt pool size, layer thickness, and CAD design.

Melt pools near boundaries significantly influence surface deviations. In the cylinder hull region, where scanning proceeds along the X-axis, melt pool depth, and distances $|OB|$, $|OD|$ dominate. In contrast, the bulk region, with alternating scan directions of 0° and 90° across even and odd layers, requires consideration of melt pool depth, $|OA|$, $|OB|$, $|OC|$, and $|OD|$ to capture inter-hatch and inter-layer dependencies. Figure 6g illustrates these dimensions, and prior work on IN625 [46] indicates melt pool depths of $\sim 90 \mu\text{m}$ in the bulk and $\sim 60 \mu\text{m}$ in the hull. Measured melt pool widths ($\sim 130 \mu\text{m}$ in the bulk and $\sim 100 \mu\text{m}$ in the hull) align closely with the sum of $|OB|$ and $|OD|$. Figure 12 summarises the melt pool depth, $|OA|$, $|OB|$, $|OC|$, and $|OD|$, with their standard deviations from section 2.3.1.

Figure 13 shows the geometrical schematic of the inner cylinder surface profile, illustrating how overlapping melt pools across layers contribute to radial deviations. Contour melt pools primarily govern deviations; however, melt pools in adjacent bulk regions also matter. The nominal melt pool was used to assess the

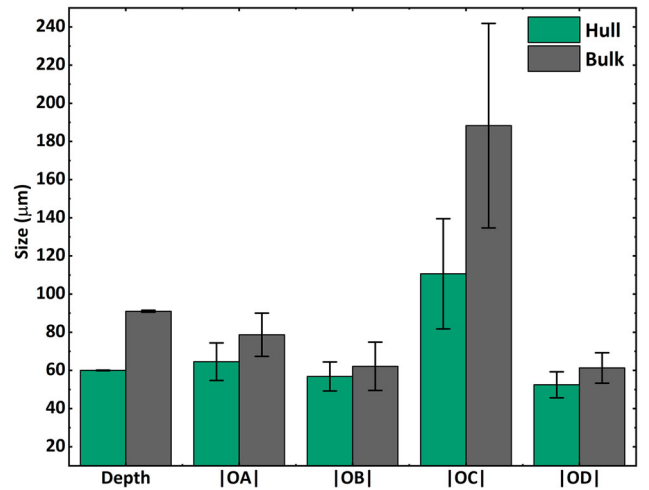


Figure 12. Melt pool geometry for hull and bulk region. $|OA|$ and $|OC|$ are the distances between the melt pool's front and back end and centroid, $|OB|$ and $|OD|$ are distances along 90° and 270° , their sum closely related to the melt pool width from the literature [46]. The melt pool depth is also obtained from the literature [46]. Error bars indicate $\pm 1\sigma$ of the standard deviation.

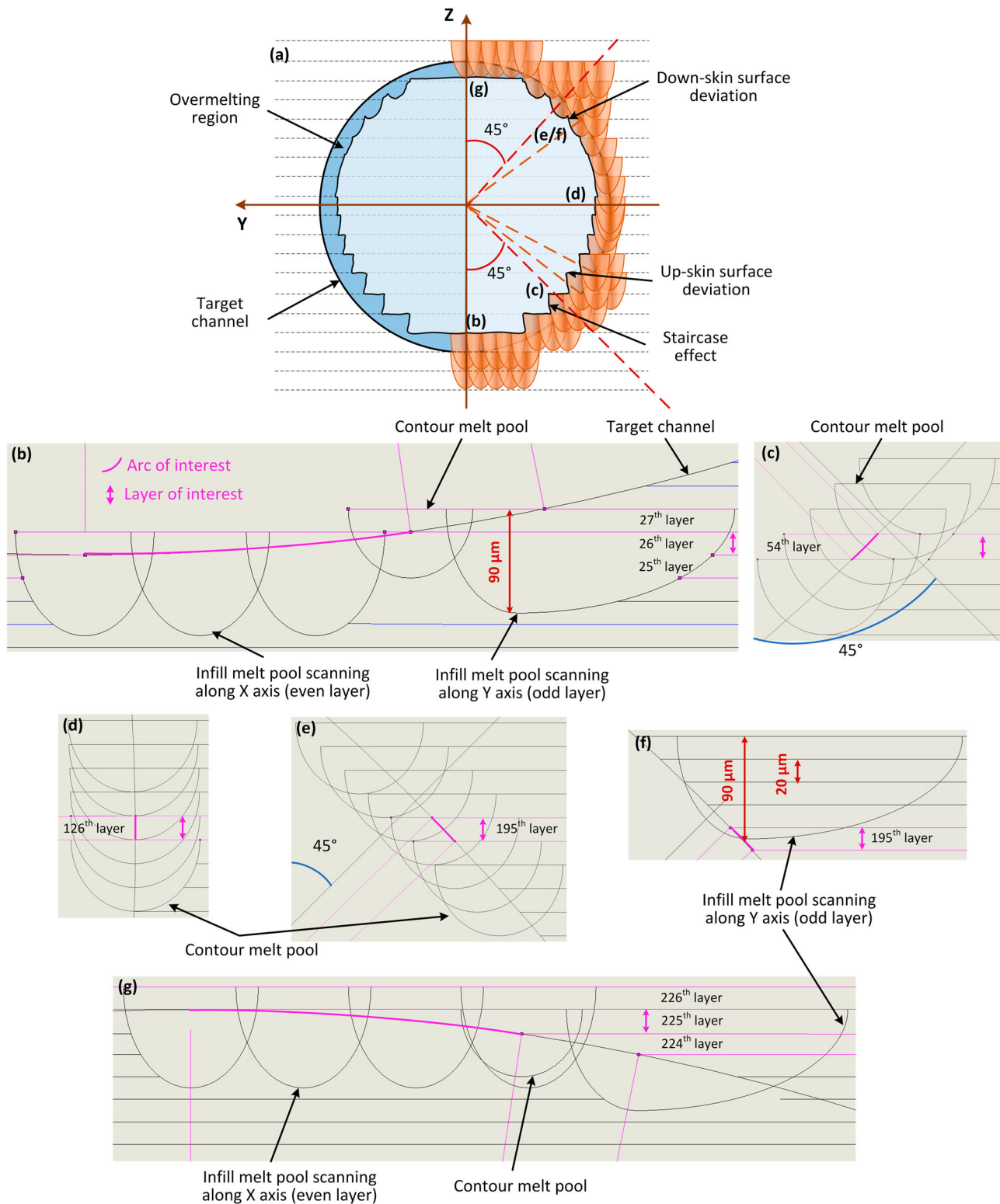


Figure 13. Geometrical schematic of (a) cylinder surface profile; (b) the 25th, 26th, and 27th layers in the cylinder bottom; the 54th (c) and 126th (d) layers in the down-skin surface; the 195th layer (e) (f) near 135°; (g) the 224th, 225th and 226th layers in the cylinder top.

impact of surrounding layers on the target layer, ignoring the melt pool variations along one scan vector. Figure 13b–g show close-ups of cylinder boundaries with melt pools.

At the cylinder bottom (Figure 13b), for example, the 26th layer's deviation approaches one layer thickness and is influenced by the 27th layer above. Near 45° (Figure 13c), the arc covered by the 54th layer is

affected by the two upper layers. In contrast, at 90° (Figure 13d), the 126th layer's arc is influenced by contour melt pools from three layers above, corresponding to the ratio of contour melt pool depth to layer thickness. However, in up-skin regions, upper-layer melt pools primarily remelt inner areas of previously solidified layers, limiting their impact on surface deviations.

In the overhang region, depicted in Figure 13e–g, arcs in the 195th and 225th layers are influenced by contour and infill melt pools. Melt pool cross-sections in the YZ plane vary between even and odd layers due to the alternating scan directions. When infilling the bulk region above the target layer, melt pools can remelt the target layer's edge. This inter-layer remelting effect, extending across four layers, is determined by the ratio of infilling melt pool depth to layer thickness. Additionally, one layer below also impacts the arc, as shown in Figure 13e.

This analysis underscores the impact of inter-layer remelting on droplet formation in overhang regions and the staircase effect on up-skin protrusions. Table 2 quantifies these effects by listing the number of layers above and below the target (0th) layer for different local orientations.

Two surrogate models, CaT_U and CaT_D, were developed for up-skin and down-skin surfaces, respectively. The numbers of layers above and below the 0th layer in Table 2 define the search space for the hyperparameters I_b (layers below) and $STEP$ (total number of layers above and below, including the 0th layer).

For up-skin surfaces ($\theta = 0^\circ$ to 90°), layers were fully supported by solidified material, minimising the influence of lower layers on the target layer. Consequently, the CaT_U model focuses on the number of layers above the 0th layer, ranging from 1 to 3. In this case, I_b was fixed at 0, and $STEP$ was searched from 2 to 4.

For down-skin surfaces ($\theta = 90^\circ$ to 180°), insufficient support from loose powder introduces significant thermal and geometric variations. The remelting effects from the layers above strongly influence the 0th layer's arc radial deviation. Additionally, the lack of adequate support leads to changes in melt pool shape,

necessitating adjustments to the inter-layer affecting ranges defined in Table 2. The layers above the 0th layer were modified to range from 2 to 6. For the CaT_D, I_b was searched from 0 to 1, and the $STEP$ was searched from 3 to 7.

The other three hyperparameters (d_m , N , d_r), which determine the CaT_U and CaT_D model scale, were limited to two configurations: (512, 6, 128) and (256, 3, 64). The two configurations provide robust baselines for comparison. Insights from initial experiments indicate that the selected configurations effectively cover the trade-offs between model complexity and performance. The hyperparameter search space for the CaT_U and CaT_D models is listed in Table 3.

3.2. Hyperparameters fine-tuning and model performance

Each model configuration was defined by a unique combination of hyperparameters I_b , $STEP$, and (d_m , N , d_r), as detailed in Table 4. The CaT_U models (indices 1–6) were evaluated using a 3-fold CV. As shown in Figure 14a, higher Transformer complexity (d_m , N , $d_r = 512, 6, 128$) improved performance, with the optimal $STEP = 3$ capturing inter-layer effects. For CaT_D models (indices 1–20), a smaller Transformer configuration (256, 3, 64) consistently outperforms the larger one, as illustrated in Figure 14b. The best results were achieved with $I_b = 1$ and $STEP = 4$. Larger $STEP$ sizes (>5) introduced noise, diminishing model performance.

The optimal configurations for CaT_U and CaT_D models are { $I_b: 0$, $STEP: 3$, (d_m , N , d_r): (512, 6, 128)} and { $I_b: 1$, $STEP: 4$, (d_m , N , d_r): (256, 3, 64)}, respectively. With these configurations, both models were retrained using 80% of the training set and validated on the remaining 20%. The training and validation loss curves are presented in Appendix B (Figure B1).

The performance metrics for the trained CaT_U and CaT_D models applied to the testing dataset are summarised in Table 5, including averages and standard deviations calculated from five runs with different

Table 2. Inter-layer remelting range for up-skin and down-skin surfaces.

Region	Local orientation θ (°)	Layers above the 0th layer	Layers below the 0th layer
Up-skin surface	0–45	1–2	0
	45–90	2–3	0
Down-skin surface	90–135	4	0–1
	135–180	4	1–0

The number of layers above and below the 0th layer was obtained from geometrical analysis.

Table 3. Search space for model hyperparameters, including the embedded feature dimension (d_m), the number of encoder blocks (N), the regressor hidden layer dimension (d_r), the number of layers below the 0th layer (I_b), and the total number of layers in sampling cuboid ($STEP$)

Hyperparameters	Model	
	CaT_U	CaT_D
I_b	0	0/1
$STEP$	2/3/4	3/4/5/6/7
(d_m , N , d_r)	(512, 6, 128)/(256, 3, 64)	(512, 6, 128)/(256, 3, 64)
Total	6	20

CaT_U and CaT_D are for the up-skin and down-skin surfaces.

Table 4. Configuration Indices for CaT_U and CaT_D Models.

CaT_U				CaT_D			
Index	l_b	STEP	(d_m, N, d_r)	Index	l_b	STEP	(d_m, N, d_r)
1	0	2	(512, 6, 128)	11	1	4	(512, 6, 128)
2	0	2	(256, 3, 64)	12	1	4	(256, 3, 64)
3	0	3	(512, 6, 128)	13	1	5	(512, 6, 128)
4	0	3	(256, 3, 64)	14	1	5	(256, 3, 64)
5	0	4	(512, 6, 128)	15	1	6	(512, 6, 128)
6	0	4	(256, 3, 64)	16	1	6	(256, 3, 64)
				17	1	7	(512, 6, 128)
				18	1	7	(256, 3, 64)
				19	1	8	(512, 6, 128)
				20	1	8	(256, 3, 64)

random seeds. These results confirm the robust performance of both models. CaT_U achieved an RMSE of $6.1 \mu\text{m}$ and an R^2 of 0.91 for the up-skin surface. For the more complex down-skin surface deviations, CaT_D demonstrated strong generalisation with an RMSE of $27.4 \mu\text{m}$ and an R^2 of 0.95. The RMSE of CaT_U corresponds to 11.3% of the average ground-truth deviation for the up-skin surface ($54.1 \mu\text{m}$), while CaT_D's RMSE represents 9.4% of the average deviation for the down-skin surface ($291.7 \mu\text{m}$).

The higher relative error for the up-skin surface is attributed to the staircase effect, a result of the discretization inherent in the layer-by-layer construction process. This effect introduces subtle surface steps that are difficult to fully capture using shape and thermal process signatures. In contrast, down-skin deviations are primarily driven by local overheating, which dominates over the influence of the staircase effect.

The reported performance metrics are based on XCT-derived labels that inherently include measurement uncertainty. To assess the model's true predictive accuracy, it is important to account for the XCT label error.

However, this error is influenced by numerous factors, including the XCT system used, scan setup, reconstruction parameters, segmentation method, and registration accuracy. Under ideal conditions, assuming optimal scan quality, working in interpolation mode, and the use of advanced surface determination algorithms, the effective error (specific to surface determination) can be reduced to approximately 1/5 to 1/10 of the voxel size (i.e. $1.2\text{--}2.4 \mu\text{m}$) [48,49].

However, $1.2\text{--}2.4 \mu\text{m}$ can be seen as too optimistic, as it only concerns surface determination and does not include all the other error sources affecting XCT measurements. Providing a fully traceable uncertainty budget for XCT measurements is currently impractical, particularly when we do not directly gather the data but instead source from trusted open datasets. Consequently, to conservatively estimate the label inaccuracy, we can use the maximum permissible error (MPE) defined by the VDI/VDE 2630-2631.3 guideline [50]. This estimation is not an uncertainty budget. It is only applicable under the conditions supported by the VDI/VDE standard; however, it is considered a valid tool for

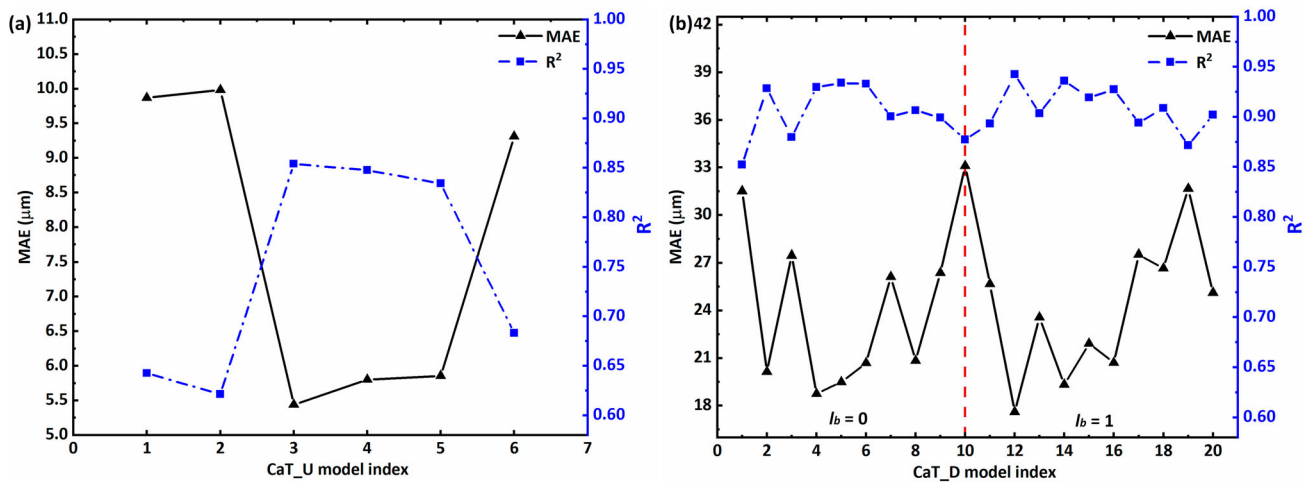


Figure 14. Hyperparameters optimisation. Mean absolute error (MAE) and R-squared (R^2) using three-fold cross-validation for (a) CaT_U and (b) CaT_D. The model index is referenced in Table 4.

Table 5. Prediction performances on the testing dataset (mean $\pm 1\sigma$ of standard deviation across five runs).

Model	Optimised configuration	RMSE (μm)	MAE (μm)	R ²
CaT_U	{ I_b : 0, STEP :3, (d_{nv} , N , d_r): (512, 6, 128)}	6.1 $\pm$ 1.4	4.1 $\pm$ 1.1	0.91 $\pm$ 0.01
CaT_D	{ I_b : 1, STEP :4, (d_{nv} , N , d_r): (256, 3, 64)}	27.4 $\pm$ 2.5	17.1 $\pm$ 1.6	0.95 $\pm$ 0.01

comparing the MPE across different XCT equipment. The parts in this study were scanned using a Zeiss METRO-TOM 800 [39]. Based on the machine's technical details [51], the MPE for length measurement is given by:

$$\text{MPE} = 9 + \frac{L}{50} \quad (3)$$

where L is the maximum measured length in mm. With a maximum feature length of 9 mm (Figure 2), this yields $\text{MPE} = 9.2 \mu\text{m}$. Although the true XCT uncertainty is likely much smaller, we adopt this value to ensure a conservative and traceable estimate for label noise/inaccuracy. This MPE-based upper bound enables us to interpret our model's true prediction performance.

The next step is to convert this MPE into a standard deviation estimate σ_{label} , which represents the typical XCT label noise. Assuming a Gaussian distribution of measurement noise, we treat MPE as a one-sided 99.7% confidence interval ($\pm 3\sigma$). Accordingly,

$$\sigma_{\text{label}} = \frac{\text{MPE}}{3} \quad (4)$$

We get $\sigma_{\text{label}} = 3.1 \mu\text{m}$. This value provides a rule-of-thumb estimate of the XCT label noise; more precise quantification will be pursued in future work.

Error propagation analysis is applied to distinguish model prediction error from measurement noise. To estimate the true performance of data-driven model:

$$\text{RMSE}_{\text{true}} = \sqrt{\text{RMSE}_{\text{obs}}^2 - \sigma_{\text{label}}^2} \quad (5)$$

where RMSE_{obs} is observed RMSE between the model prediction and the XCT label (which includes measurement noise). $\text{RMSE}_{\text{true}}$ is the RMSE between the model prediction and the true surface deviation.

Based on Eq. (5), when the observed prediction error approaches the XCT measurement uncertainty, the estimated true model error becomes negligible, which is an outcome that may not be realistic. In such cases, the model may appear accurate, but this apparent accuracy could stem from fitting fluctuations within the noise envelope rather than capturing true surface deviations.

In this study, the observed RMSEs of 6.1 μm (up-skin) and 27.4 μm (down-skin) exceed the estimated XCT label noise, suggesting that the CaT_U and CaT_D models are capturing real, structured surface deviations rather than

fitting measurement noise. The corresponding true model errors ($\text{RMSE}_{\text{true}}$) are calculated as 5.3 and 27.2 μm , respectively, supporting the model's high fidelity in resolving deviation patterns.

Meanwhile, our approach will continue to accept XCT data with noise as ground truth, report the RMSE relative to this data, and acknowledge that integrating XCT uncertainty into the RMSE remains a critical direction for future investigation.

3.3. Comparing with other models

In the proposed CaT model, a 2D-CNN and a Transformer encoder are connected sequentially. The 2D-CNN extracts low- to mid-level spatial features from each layer within the sampling cuboid, while the Transformer encoder employs a self-attention mechanism to capture inter-layer spatial dependencies. This hybrid design leverages the strengths of both localised feature extraction and global contextual reasoning, enabling more accurate predictions of surface deviations. The architecture is particularly well-suited for datasets that exhibit both spatial complexity and temporal (layerwise) dynamics.

To benchmark the CaT model's performance, three alternative architectures were evaluated: 3D ResNet [47,52], Swin3D [53], and ConvLSTM. The 3D ResNet applies volumetric convolutions across the X, Y, and Z dimensions, allowing the model to capture local and global patterns across multiple scan tracks and layers. Swin3D segments the input volume into 3D patches and uses window-based self-attention, with shifted attention windows to capture long-range interactions throughout the cuboid. In the ConvLSTM baseline, the Transformer encoder in CaT is replaced by a Long Short-Term Memory (LSTM) network to model inter-layer temporal dependencies, as described in Appendix C (Table C1). These models collectively serve as comprehensive baselines for analysing multi-track, multi-layer in-process monitoring data.

Performance comparisons, shown in Figure 15, highlight that CaT consistently outperforms other CNN-only models for both up-skin and down-skin surfaces. In the PBF-LB/M process, the sequential layering of material necessitates models that can leverage historical context, where prior layers provide the initial state that is continually updated with newly deposited layers. Sampling cuboids span multiple layers, and sequential models (CaT and ConvLSTM) excel by processing them in two stages: layerwise 2D convolution for spatial features and 1D temporal reasoning for inter-layer dynamics. This architecture effectively captures dependencies between scan tracks and layers, particularly

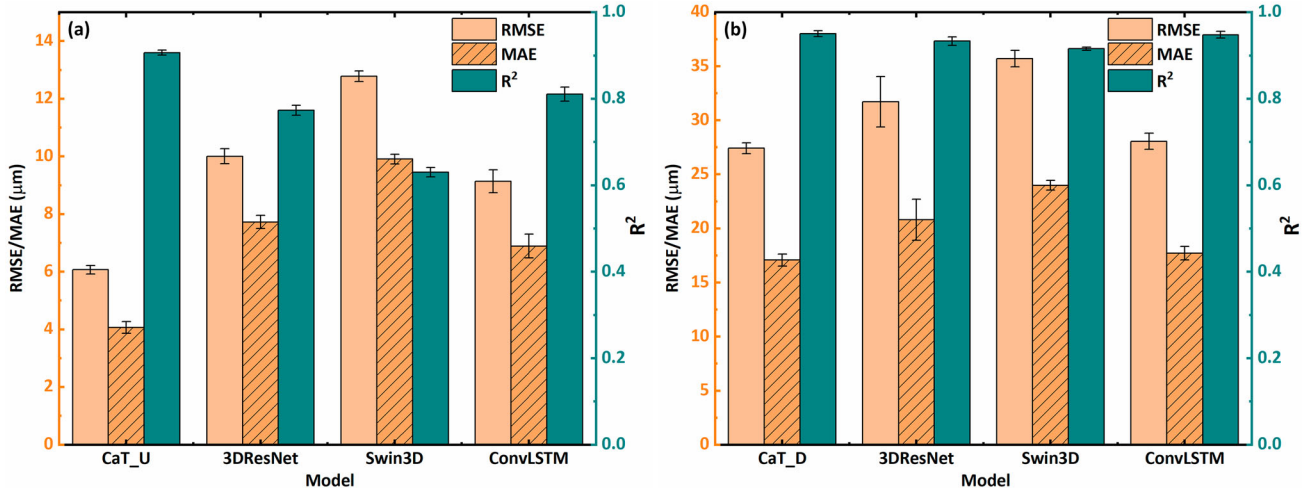


Figure 15. Performance comparison of different models: (a) up-skin and (b) down-skin surfaces.

sequential context, improving generalisation and prediction accuracy. In contrast, 3D ResNet and Swin3D process the input cuboid as a single 3D volume. While this enables effective spatial reasoning, these models may overlook critical inter-layer temporal correlations.

Besides, the CaT model achieves an average inference latency of 425 ms per prediction, making it suitable for potential in-process feedback control. Further details are provided in Appendix D.

3.4. Ablation experiments

This study highlights combining shape and thermal-related gray value features in predicting surface deviations. An ablation study was conducted to evaluate the individual and combined contributions of these feature types, with results summarised in Table 6.

The results indicate that shape and thermal features play complementary roles, with their relative importance varying across different surface regions. Shape features are especially influential in the up-skin areas, where surface deviations arise from geometric accumulation. The boundary material is not remelted by subsequent layers (Figure 13b,c) for these up-skin regions. In

contrast, down-skin deviations are primarily driven by local overheating and dross formation caused by inter-layer remelting. Thermal features effectively capture these effects by encoding heat-related anomalies and melt-solidification dynamics (Figure 13f,g).

Combining both feature types consistently yields the best performance, enabling the model to integrate geometric precision from shape descriptors with thermal context from gray value features. This fusion enhances the model's ability to generalise across varying surface conditions, contributing to high-fidelity predictions of surface deviations.

3.5. Model interpretation

Data-driven models are often referred to as 'black boxes' due to their opaque feature learning and decision-making processes, which complicate trust and interpretability. To address this challenge, model-specific interpretation techniques were employed. First, we examined feature importances and analyzed CNN filters, recognising that while the CaT model accurately predicts surface deviations, its deeper feature maps remain challenging to interpret directly. Grad-CAM was therefore applied to highlight the input regions most critical to the model's decisions. We then explored inter-layer dependencies by visualising the attention weights of the Transformer encoder, interpreting how layers influence each other from different perspectives.

Shape (42 channels) and thermal (3 channels) sampling cuboids were processed through hierarchical convolution layers, each representing one feature. To identify the most predictive channels, the 64 filters in the first CNN layer, which directly interact with the input data, were analysed. Channel importance was quantified by calculating the average absolute weights of

Table 6. Ablation study results evaluating the contributions of shape and thermal features, with metrics reported as RMSE, MAE, and R^2 metrics.

Region	Shape feature	Thermal feature	RMSE (μm)	MAE (μm)	R^2
Up-skin	✓	✓	6.1 ± 1.4	4.1 ± 1.1	0.91 ± 0.01
	✓	×	8.2 ± 1.7	6.4 ± 0.8	0.85 ± 0.03
	×	✓	9.5 ± 0.8	7.2 ± 0.7	0.79 ± 0.03
Down-skin	✓	✓	27.4 ± 2.5	17.1 ± 1.6	0.95 ± 0.01
	✓	×	34.6 ± 1.8	26.8 ± 1.1	0.87 ± 0.01
	×	✓	33.6 ± 1.6	24.7 ± 1.8	0.89 ± 0.01

Metrics are reported as mean $\pm 1\sigma$ over five runs.

these filters for each input channel, following established methodologies for interpreting early CNN layers [54–56].

The top three shape features (**dis90**, **dis260**, **dis270**) predominantly affect the up-skin region (Figure 16a), while (**dis30**, **dis210**, **A_m**) drive down-skin surfaces (Figure 16b). Shape features contribute more to up-skin predictions than thermal features, underscoring the importance of tracking the liquid melt pool boundary for predicting fully supported layer edges. In contrast, three thermal features have a higher average importance in down-skin areas, highlighting the role of thermal history in powder sintering at the overhang regions.

After identifying the most critical features, the next step is to examine how CNN's filters interact with these features. To illustrate filter weight distributions, Figure 17 shows six of the 64 filters from CNN's first layer, where each filter contains multiple channels corresponding to different features. Specifically, we focus on three gray-value channels (**T_T**, **M_T**, **A_T**) and the key shape channels (**A_m**, **dis30**, **dis210** for down-skin, and **dis90**, **dis260**, **dis270** for up-skin). Each row in Figure 17 represents a filter, while each column corresponds to a specific feature or channel. Dark squares indicate smaller weights and lighter squares signify larger weights.

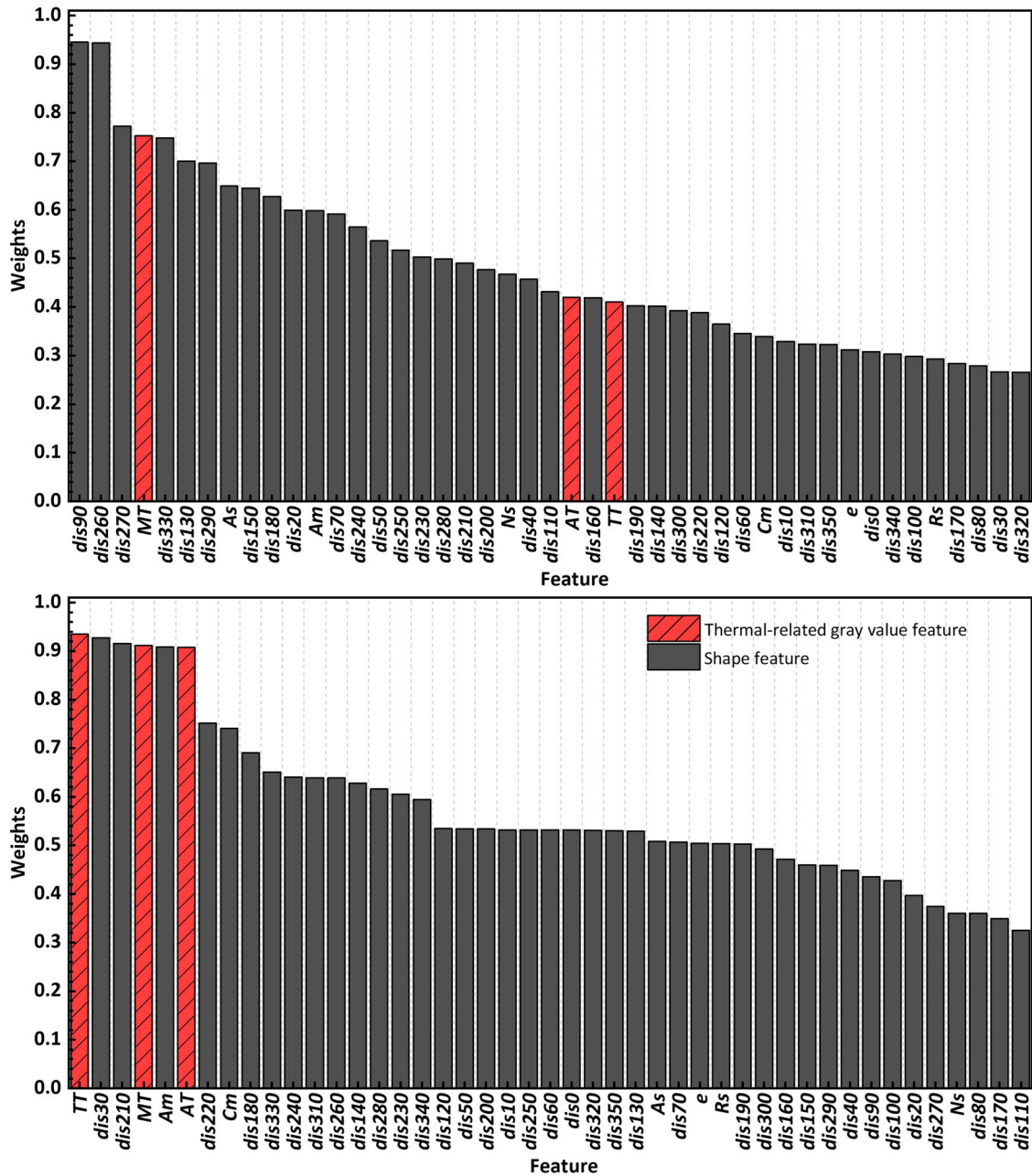


Figure 16. Feature importance analysis for (a) up-skin and (b) down-skin surface.

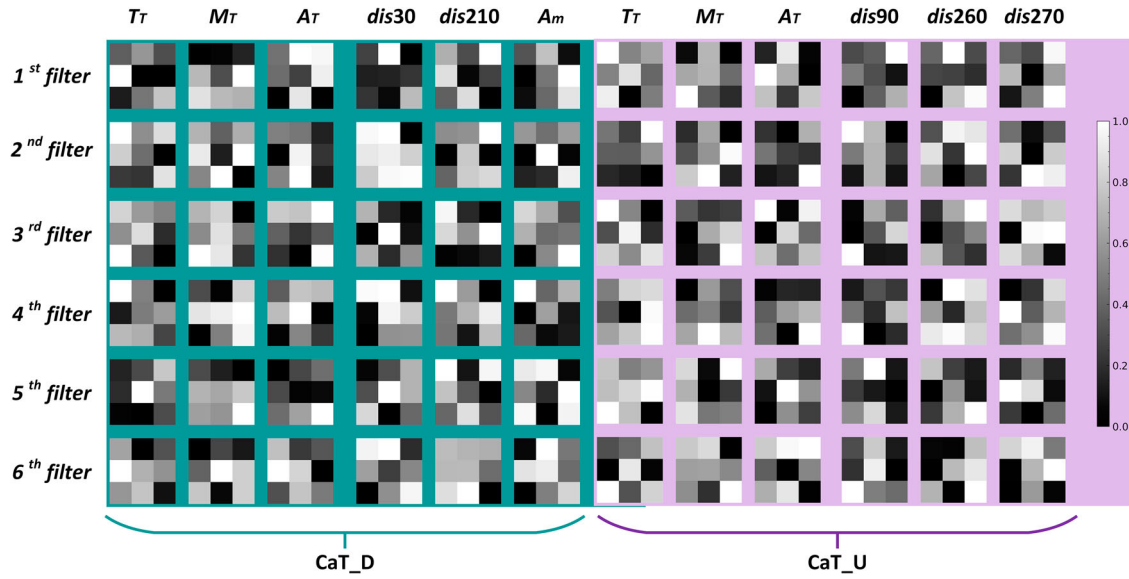


Figure 17. Visualisation of the first six filters from the first convolution layer of CaT_U and CaT_D models, each column representing a specific channel of interest. Dark squares indicate small weights, while light squares denote large weights.

These learned filters perform various tasks, including edge detection, blurring, and identifying hotspots and coldspots. Some mimic Gaussian blurring to reduce noise, while others resemble Sobel filters for edge detection. This diversity allows the CaT model to extract critical spatial and thermal features, enhancing its ability to capture nuanced data patterns.

To illustrate the impact of CNN filters, a thermal sampling cuboid was processed through the first four convolutional layers (Figure 18). The learned feature maps from the initial six filters at each layer highlight how the model evolves from detecting basic lines, bulk regions, or hull areas in the first convolutional layer to focusing on more generalised inter-hatch patterns in deeper layers. This hierarchical transformation begins with basic regional or directional features and abstracts them into complex representations, ultimately forming higher-level concepts critical for accurate predictions.

However, mapping these deeper feature representations directly to physical phenomena remains challenging. Grad-CAM provides another perspective by computing the gradient of the negative regression loss concerning the final CNN layer's activations, thereby generating localisation maps. Therefore, we can highlight intra-layer regions most influential to the CaT's predictions (Figure 19).

The most informative shape and gray-value channels are *dis90* and *M_T* for the up-skin region and *dis30* and *T_T* for the down-skin region. These channels' grayscale images and corresponding positions within the cuboids are shown in Figure 19b,d,g,i. CNN's focus varies significantly between the two areas. The up-skin surface

(Figure 19e,j) prioritises the contour and adjacent regions, emphasising boundaries where contour melt pools dominate. For the down-skin surface (Figure 19c, h), the attention is distributed more broadly across boundary and bulk regions. This broad focus captures localised boundary and bulk material behaviour, critical for down-skin areas where limited support leads to heat accumulation, over-melting, and dross formation.

To conduct quantitative validation of Grad-CAM maps, we performed a statistical correlation analysis between the average Grad-CAM activation intensities and the corresponding XCT-measured deviations within each sampling cuboid. The results revealed statistically significant positive correlations (Pearson's correlation coefficients: 0.68 for up-skin and 0.74 for down-skin), confirming that regions highlighted by Grad-CAM correspond to areas of larger surface deviation.

To further understand the Transformer encoder's role in capturing inter-layer dependencies, we analysed the self-attention mechanism. This mechanism extracts interactions among **STEP** layers within a cuboid. Analysing the attention weight matrix reveals how the model exchanges information between layers. The attention matrix, shaped as **STEP** × **STEP**, provides attention logits for each layer relative to all other layers. The column corresponding to the 0th layer, the layer of interest, was transposed and analysed in Figure 20. The CaT_U model contains six attention blocks with eight heads each, while CaT_D has three blocks with the same eight-head configuration.

Analysis of attention matrices across all Transformer heads and blocks reveals distinct patterns in how the target layer attends to its neighbouring layers. For the

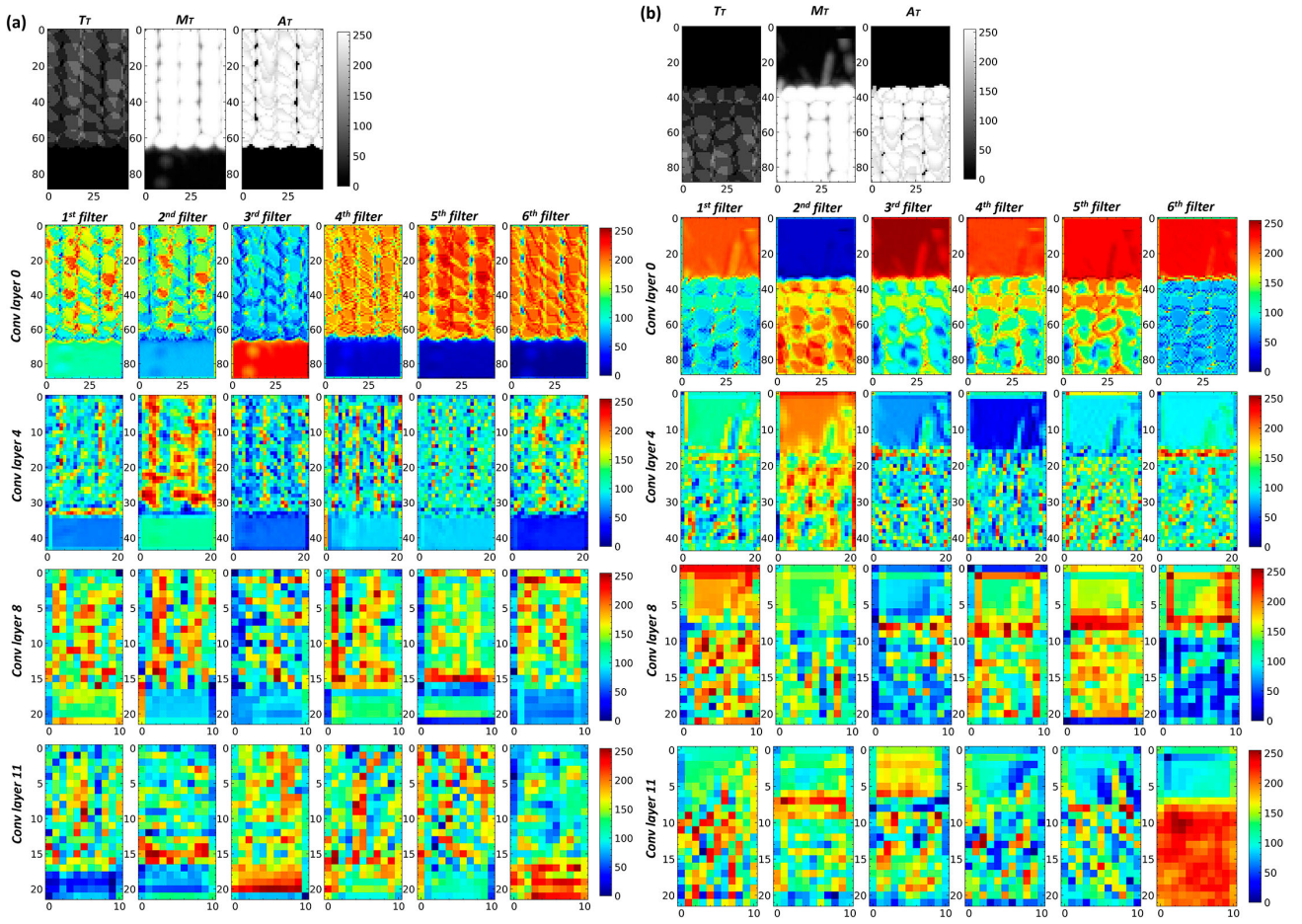


Figure 18. Visualisation of a thermal sampling cuboid and the learned patterns from the first four convolution layers of (a) CaT_U and (b) CaT_D. The first six filters of each convolution layer were analysed.

up-skin surface (Figure 20a), 43% of heads assign dominant attention to the target (0th) layer, while 55% focus on the 2nd (last) layer. This indicates that the initial melt condition and remelting from the final layer in the cuboid highly influence the prediction. For the down-skin surface (Figure 20b), 50% of heads give the greatest attention to the 2nd layer and 42% to the -1st (first) layer; attention to the 0th and 1st layers appears in about 25% of heads. Because a single head can attend to multiple layers, totals can exceed 100%. These patterns show that, in down-skin regions, both the earliest and latest layers in the sampling cuboid are critical for capturing surface deviation.

These findings underscore the importance of selecting an appropriate **STEP** size and deciding whether to include the -1st layer when modelling inter-layer remelting effects.

3.6. Model generalizability

To evaluate generalizability, the CaT models were tested on a 45° overhang surface, a geometry

distinct from the cylinder hole used in the training stage. The aim was to assess whether the CaT_U and CaT_D models could accurately predict surface deviations in a different geometric context. The performance metrics for both models are presented in Table 7.

The CaT_D model, designed for down-skin surfaces, demonstrates stronger predictive accuracy in the overhang region. It reflects its ability to capture the complex interplay of thermal and geometric factors in limitedly supported areas prone to over-melting and dross formation.

By contrast, the CaT_U model, optimised for up-skin surfaces, exhibits limitations when applied to overhangs. The reduced performance of CaT_U on the 45° overhang is likely due to geometric mismatch, as the up-skin model was trained on fully supported horizontal surfaces but evaluated on unsupported, angled geometries with different thermal boundary conditions. This suggests that incorporating a more diverse range of geometries into the training set could significantly improve generalisation. These findings emphasise the

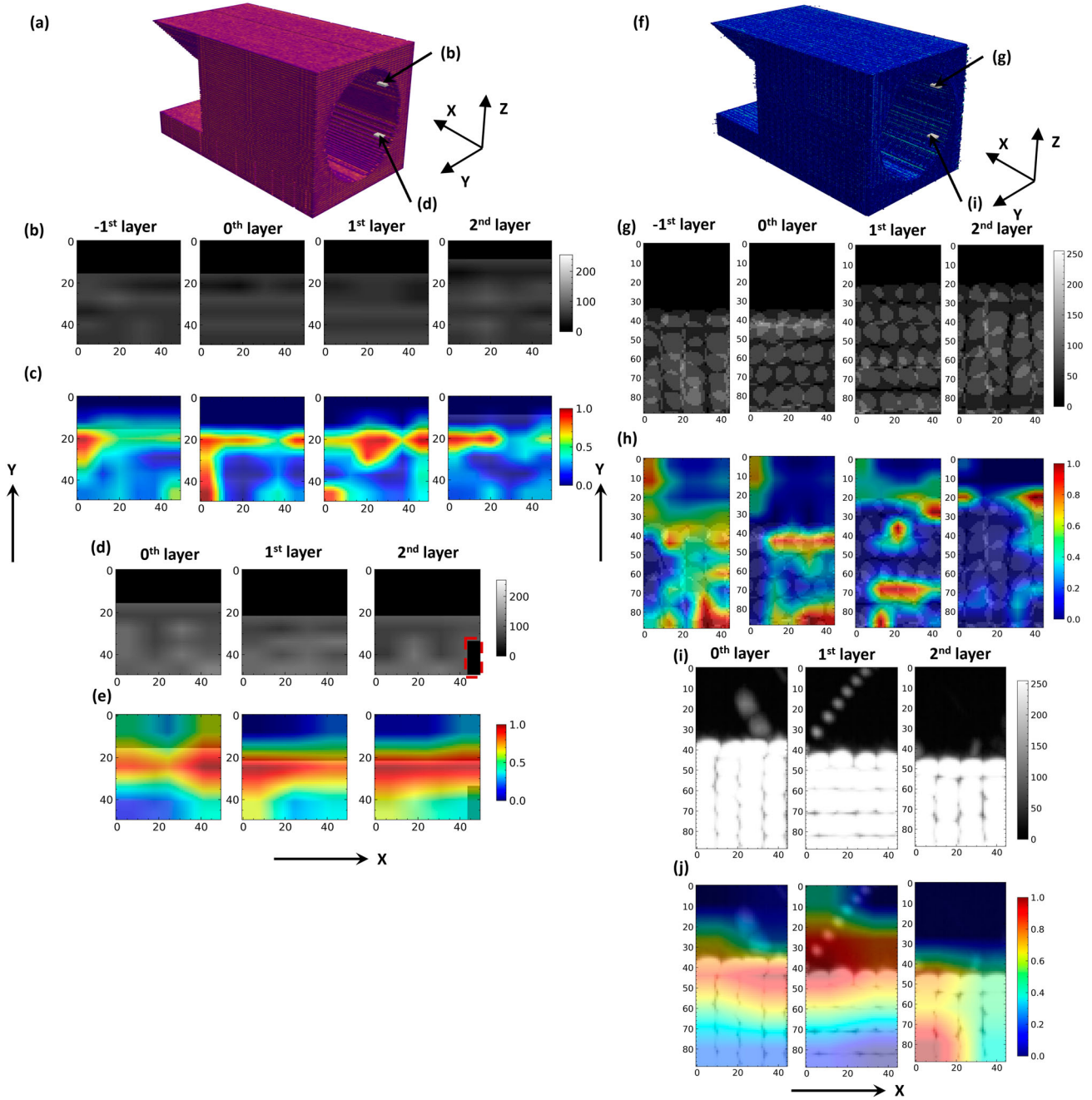


Figure 19. Visualisation of essential regions for CNN prediction using Gradient-weighted class activation mapping (Grad-CAM): (a) and (f) show sampling cuboids from the shape and gray-value feature volumes; here, the cuboid for the up-skin region includes three layers, with two layers above the 0th layer, while the down-skin cuboid covers four layers, with one layer below and two layers above the 0th layer. (b) and (g) represent cuboids of dis_{30} and T_T in the down-skin region; (d) and (i) are cuboids of dis_{90} and M_T in the up-skin region; (c), (e), (h), and (j) display the Grad-CAM results, with red areas highlighting regions motivating the final prediction. The black area within the dashed red frame in (d) is due to the misalignment between the sampling distance and hatch space, as discussed in Figure 7.

importance of fine-tuning data-driven models to the specific characteristics of the analysed regions. We also acknowledge that the current results are specific to the tested geometry and material. Formal deployment would require additional steps such as external validation or certification.

3.7. Comparison to previous research

We benchmark our approach against recent studies that utilise the NIST public dataset or related builds, as listed in Table 8. Jung et al. [57] utilised a k-nearest neighbour (k-NN) model to classify energy density levels based on melt pool geometry features, contributing to parameter

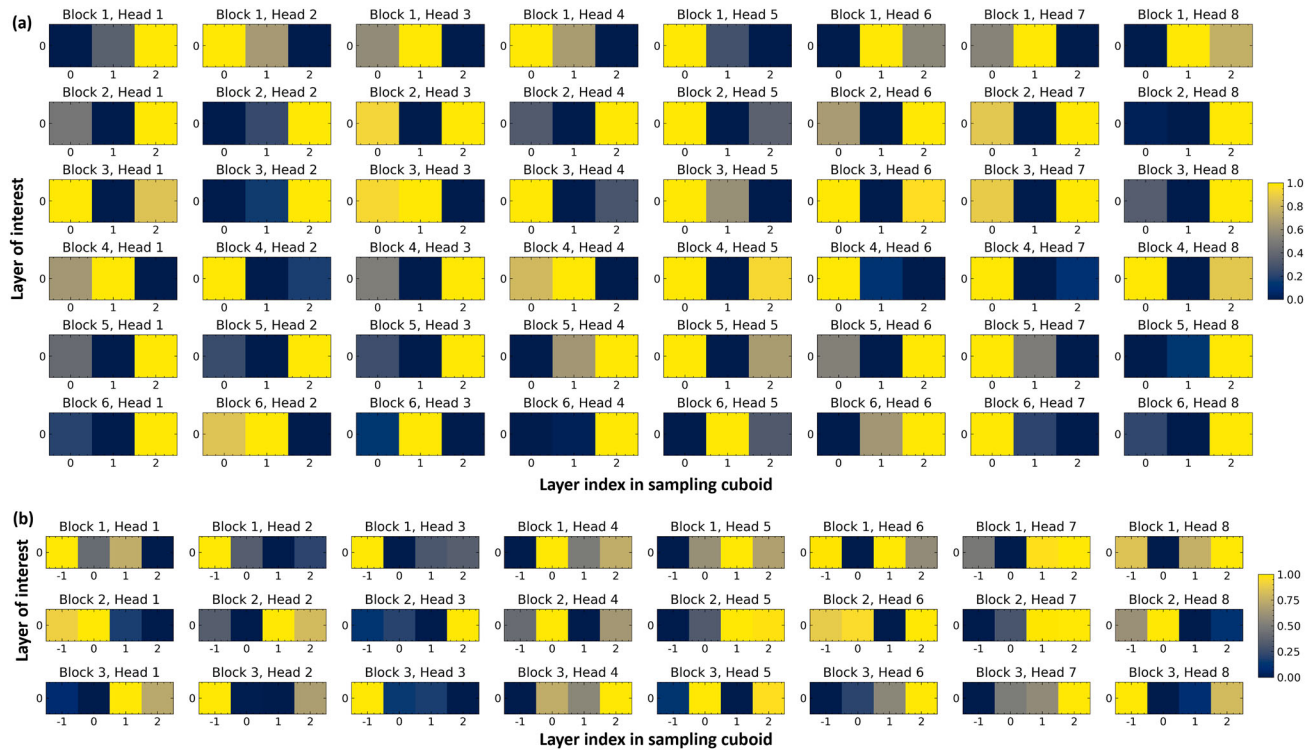


Figure 20. Visualisation of the attention weight matrix for the 0th layer. The heatmaps present attention scores from the 0th layer to all layers in the sampling cuboid for (a) CaT_U and (b) CaT_D. The layer index 0 is the 0th layer, while -1 means the layer below the 0th layer.

Table 7. Performance metrics for the CaT_U and CaT_D models evaluated on the 45° overhang surface and horizontal cylinder regions

Model	Region	RMSE (μm)	MAE (μm)	R^2
CaT_U	Horizontal cylinder (Up-skin)	6.1 ± 1.4	4.1 ± 1.1	0.91 ± 0.01
	45° Overhang	25.4 ± 0.3	17.1 ± 0.5	0.83 ± 0.01
CaT_D	Horizontal cylinder (Down-skin)	27.4 ± 2.5	17.1 ± 1.6	0.95 ± 0.01
	45° Overhang	10.3 ± 1.4	8.1 ± 0.9	0.92 ± 0.02

inference. Williams & Sing [58] developed a convolutional recurrent neural network to detect overhang regions using MPM images. They demonstrated high accuracy ($>99\%$) for binary classification. Chiappetta et al. [59] focused on uncertainty quantification for

residual strain predictions in thermomechanical simulations of an Inconel 625 cantilever beam. While their Bayesian inversion framework improved the estimation of simulation parameters (e.g. convection coefficients), their focus is on post hoc validation of numerical

Table 8. Comparison of our study with recent works using NIST open datasets.

Study	Dataset	Objective	Input Data	Model Type	Output & Resolution	Scope & Generalizability
Jung et al. [57]	20180708-HY-3D Scan Strategies [38]	Energy density classification	Melt pool geometry features	k-NN classifier	Discrete class label (91.4% accuracy)	Multiple scan strategies on near rectangular prisms
Williams & Sing [58]	Overhang Part X4 [33] and Overhang Part X16 [60]	Surface quality classification	Co-axial MPM image stacks	Conv-Recurrent Neural Network	Binary label (overhang vs. no-overhang), $> 99\%$ accuracy	Horizontal channels
Chiappetta et al. [59]	AMBench2018-01 [61]	Simulation uncertainty quantification	Displacement experimental data	Multi-index stochastic collocation method	Uncertainty quantification of the residual strains prediction	Post hoc validation, no in-situ prediction
This study	Overhang Part X4 [33]	Surface deviation prediction	melt pool geometry and thermal-related gray value features	CNN-Transformer hybrid	Regression of actual-to-nominal deviation, spatially continuous, CAD-aligned	Horizontal channels and 45° overhang surfaces

simulations, not direct prediction from in-situ monitoring. In contrast, our study focuses on the regression of geometric deviations in complex geometries, aiming to predict surface quality. We develop a data-driven surrogate that maps co-axial sensing data to actual-to-nominal labels, demonstrating its potential for feedback control and geometry-informed design optimisation.

In sum, these studies highlight the relevance and good value of the NIST dataset. However, none of them provide localised, supervised predictions of internal surface deviations directly from in-process monitoring data. Our work uniquely integrates geometric context, gray-value thermal history, and high-resolution XCT labelling into a unified surrogate framework.

4. Conclusions

This study introduced a high-fidelity surrogate model for predicting surface deviations in metallic laser powder bed fusion (PBF-LB/M). By extracting 42 shape features and 3 thermal-related gray value features from melt pool monitoring images and mapping them to CAD geometry, 3D voxelated feature volumes were generated. Surface deviations were characterised using X-ray computed tomography, forming actual-to-nominal error maps. Sampling cuboids moving through these feature volumes and the corresponding segmented error labels served as supervised datasets for the model training.

This surrogate model combines a 2D Convolutional Neural Network (CNN) with a Transformer Encoder (CaT) to capture the intricate spatio-temporal relations inherent in the PBF-LB/M. More specifically, the CNN extracts localised features from each layer of the sampling cuboid. The Transformer encoder then processes the sequence of layerwise features through its self-attention mechanism to model inter-layer remelting. The gradient-weighted class activation mapping reveals that the CNN focuses on boundary regions, aligning with empirical strategies for predicting surface deviations. The self-attention maps from the Transformer encoder highlight critical layers within the sampling cuboid, structuring high-level output features to refine predictions for the target layer's surface deviation.

The CaT model was tailored for up-skin (CaT_U) and down-skin (CaT_D) surfaces in the internal cylinder regions of the test parts. CaT_U achieved an RMSE of 6.1 μm , an MAE of 4.1 μm , and an R^2 of 0.91, with the RMSE representing 11.3% of the average 54.1 μm ground-truth deviation for the up-skin surface. Similarly, CaT_D performed well, with an RMSE of 27.4 μm , an MAE of 17.1 μm , and an R^2 of 0.95, relative to 9.4% of the 291.7 μm average deviation for the down-skin surface.

Testing these models on the 45° overhang surface confirmed their robustness and generalizability.

The proposed model enables two key applications: (1) feedback control by mapping predicted deviations to guide adaptive scan parameter adjustments and (2) design optimisation by identifying geometry-sensitive regions requiring orientation or support refinement. Interpretability, as enhanced by Grad-CAM and attention analysis, improves traceability, facilitating industrial deployment and future qualification efforts.

In summary, this study establishes that high-fidelity surface deviation predictions in PBF-LB/M rely on three key factors: measuring process signatures sensitive to surface deviations, constructing a high-quality labelled dataset, and considering inter-hatch and inter-layer effects in the modelling process. The methodologies and insights presented here advance the predictive modelling of surface deviations in PBF-LB/M, emphasising the critical role of integrating multidimensional process signatures with advanced machine learning architectures. Additionally, these findings suggest that future developments in PBF-LB/M process monitoring should focus on capturing and modelling the complex interdependencies between scan tracks and layers to achieve accurate predictions of as-built surface quality.

While the proposed model demonstrates strong predictive performance and generalisation across surface types (e.g. up-skin, down-skin, and 45° overhang) within a single material system (IN625), its applicability to other materials, more complex geometries, and varied process parameters remains an important direction for future work. Planned extensions include validation on additional alloys such as Ti-6Al-4 V, SS316L, and copper alloy, as well as builds featuring multi-curvature channels and internal lattice structures. These efforts will aim to assess model transferability and enhance robustness across various machines, materials, and building designs. Besides, integrating XCT uncertainty into the performance evaluation remains a critical direction for future investigation.

Author contributions

Zhengrui Tao: Writing – original draft, Software, Investigation, Formal analysis, Visualisation. **Mirko Sinico:** Formal analysis, Investigation, Visualisation, Writing – review & editing. **Bey Vrancken:** Investigation, Writing – review & editing. **Wim Dewulf:** Conceptualisation, Funding acquisition, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This research was funded by the EU Framework Programme for Research and Innovation - Horizon 2020 – [grant number 956172] within the xCTing research project..

Data availability statement

The data that support the findings of this study are available from the corresponding author, Z. Tao, upon reasonable request. The code of this research has been saved on a storage site, whose website is <https://doi.org/10.48804/RRQVDH>.

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